

Sid Mathur's Thematic ETF Screener – Daily Report

2026-07-07

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Action Center**Run date:** 2026-07-07 **Universe:** 660 ETFs / 26 clusters**Full PC1:** FLAT **Clusters long:** 9/26 **ETFs long:** 282/660 (43%)**Today's events (cluster level)**

spread_buy = 5dma – 50dma; spread_sell = 5dma – 20dma (price units). A BUY fired today iff spread_buy crossed above 0; a SELL iff spread_sell crossed below the cushioned threshold.

BUY (1)

cluster	theme	spread_buy	p_h5	nss
cluster_4	Developed ex-US Equity	0.3	0.3	-0.272

SELL (1)

cluster	theme	spread_sell	p_h5	nss
cluster_26	Leveraged / Inverse	-0.304	0.74	-0.492

Imminent (cluster level)

Ranked by short-horizon NSS (h=1 day) so candidates that could trigger *tomorrow* surface above those that need several days; p_h5 is the 5-day probability for context. **spread_buy = 5dma – 50dma, spread_sell = 5dma – 20dma** (price units; closer to 0 = closer to crossover). **Imminent BUYs are restricted to clusters whose 252d trend filter currently approves** – otherwise the cross would be blocked at entry, so the imminence is moot.

Imminent BUY (top 5 by h=1 NSS, filter-approved)

cluster	theme	spread_buy	p_h1	p_h5	nss_h1	nss_h5
cluster_7	U.S. Small Cap Equity	5.319	1.000	1.00	1.000	1.00
cluster_1	U.S. Mid Cap Equity	5.471	1.000	1.00	1.000	1.00
cluster_15	Real Estate	2.029	1.000	0.99	0.982	0.90
cluster_5	U.S. Large Cap Equity	1.787	0.980	0.48	0.928	0.18
cluster_3	U.S. Large Cap Growth	3.274	0.977	0.42	0.777	-0.09

Imminent SELL (top 2 by h=1 NSS)

cluster	theme	spread_sell	p_h1	p_h5	nss_h1	nss_h5
cluster_19	Broad Commodities	-6.993	1.000	1.00	-1.000	-1.00
cluster_25	International Fixed Income	-0.349	0.999	0.93	-0.999	-0.93

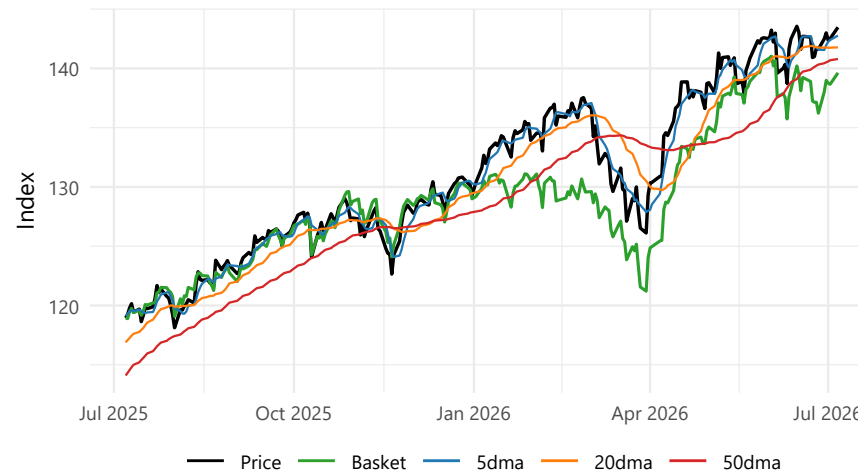
Full Sample PC1

Figure 1: Full sample PC1 portfolio price with short / sell / buy moving averages and replication-basket overlay (last 12 months). Basket rescaled to start at PC1's value at the left edge of the visible window.

Replication basket: VOO basket cor: 0.94 sticky

Current state: FLAT

5dma: 142.763 20dma: 141.772 50dma: 140.791

Spread vs 50dma: +1.972 Spread vs 20dma: +0.991

P(buy crossover within 5 days): 86.2% P(sell): 17.3%

Position Sheet

Replicate each currently-long cluster via its 1-5 member basket. Weights are NNLS (non-negative least squares) – long-only and tradeable – and sum to ~1 per cluster for unit-variance replication. Scale each cluster's basket to your target cluster allocation.

Active positions (long clusters)

ticker	cluster	theme	nss	days	return	pos_wt	cor
VHT	cluster_12	Healthcare	1.000	40	13.3%	8.15%	0.93
MUB	cluster_16	Municipal Bond	0.997	25	0.5%	18.09%	0.95
VTV	cluster_2	U.S. Large Cap Value	0.996	57	5.9%	14.66%	0.98
XLP	cluster_18	Consumer Staples	0.949	5	-0.9%	5.58%	0.99
XLU	cluster_18	Consumer Staples	0.949	5	-0.9%	3.45%	0.99
BND	cluster_6	Aggregate Fixed Income	0.213	6	-0.4%	14.58%	0.99
TLT	cluster_14	Fixed Income	0.171	13	-0.4%	12.66%	0.97
VEA	cluster_4	Developed ex-US Equity	-0.272	0	0.0%	7.24%	0.99
BWX	cluster_25	International Fixed Income	-0.926	24	-2.2%	15.02%	0.93
PDBC	cluster_19	Broad Commodities	-1.000	21	-13.3%	5.55%	0.98

Watchlist (imminent-buy clusters, $p > 80\%$, not yet long)

ticker	cluster	theme	p_h5	nss	weight	cor
IJH	cluster_1	U.S. Mid Cap Equity	1.00	0.997	1.004	0.98
IJR	cluster_7	U.S. Small Cap Equity	1.00	1.000	0.930	0.98
VNQ	cluster_15	Real Estate	0.99	0.901	0.970	0.99
EWJ	cluster_22	Japan Equity	0.82	0.359	0.910	0.97

Cluster Overview

name is derived from the basket members' categories (e.g. "Technology" or "Tech + Financials"). change shows the LONG/FLAT shift vs the prior run: →LONG or →FLAT for a state flip, new if this cluster had no match in yesterday's bundle, shifted if the cluster's identity changed (top-3 members by |PC1 weight| no longer match), empty if state is unchanged. dNSS is today's NSS minus the prior run's.

	id	name	category	n	signal	change	days	return	nss	dNSS
7	cluster_7	U.S. Small Cap Equity	U.S. Small Cap Eq...	42	FLAT	shifted			1.000	+0.440
12	cluster_12	Healthcare	Healthcare	17	LONG		40	13.3%	1.000	+0.000
1	cluster_1	U.S. Mid Cap Equity	U.S. Mid Cap Equi...	74	FLAT	shifted			0.997	-0.003
16	cluster_16	Municipal Bond	Municipal Bond	13	LONG		25	0.5%	0.997	-0.002
2	cluster_2	U.S. Large Cap Value	U.S. Large Cap Va...	73	LONG	shifted	57	5.9%	0.996	+0.043
18	cluster_18	Consumer Staples	Consumer Staples	11	LONG		5	-0.9%	0.949	-0.031
15	cluster_15	Real Estate	Real Estate	14	FLAT				0.901	-0.063
22	cluster_22	Japan Equity	Japan Equity	9	FLAT				0.359	+0.286
6	cluster_6	Aggregate Fixed Income	Aggregate Fixed I...	43	LONG	shifted	6	-0.4%	0.213	+0.692
5	cluster_5	U.S. Large Cap Equity	U.S. Large Cap Eq...	46	FLAT				0.182	+0.541
14	cluster_14	Fixed Income	Fixed Income	15	LONG	shifted	13	-0.4%	0.171	-0.829
20	cluster_20	India Equity	India Equity	10	FLAT				-0.063	-0.005
3	cluster_3	U.S. Large Cap Growth	U.S. Large Cap Gr...	67	FLAT	shifted			-0.089	+0.681
4	cluster_4	Developed ex-US Equity	Developed ex-US E...	60	LONG	shifted	0	0.0%	-0.272	-1.019
26	cluster_26	Leveraged / Inverse	Leveraged / Inver...	2	FLAT				-0.492	-0.295
24	cluster_24	Cash / Ultra-Short Fixed In...	Cash / Ultra-Shor...	9	FLAT				-0.501	-0.584
8	cluster_8	Asia-Pacific Equity	Asia-Pacific Equi...	25	FLAT				-0.556	+0.168
13	cluster_13	China Equity	China Equity	15	FLAT				-0.700	+0.209
9	cluster_9	Emerging Markets Equity	Emerging Markets ...	25	FLAT				-0.767	+0.098
11	cluster_11	High Yield Bond	High Yield Bond	21	FLAT				-0.815	+0.098
17	cluster_17	Gold	Gold	12	FLAT				-0.880	+0.078
25	cluster_25	International Fixed Income	International Fix...	6	LONG		24	-2.2%	-0.926	+0.048
21	cluster_21	Inflation-Protected Fixed I...	Inflation-Protect...	10	FLAT				-0.952	-0.104
23	cluster_23	Bank Loan / Floating Rate	Bank Loan / Float...	9	FLAT				-0.997	+0.003
10	cluster_10	Materials	Materials	21	FLAT				-0.998	+0.002
19	cluster_19	Broad Commodities	Broad Commodities	11	LONG		21	-13.3%	-1.000	+0.000

Cluster Details

Cluster 1 – U.S. Mid Cap Equity

Members: 74 **Var explained:** 0.846 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-13 – filter **PASS** (252d at date = +30.6%)
- Latest sell_cross: 2026-05-19
- Current 252d return: +23.1% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-19 more recent than last effective buy 2026-04-13)



Category breakdown

category	n_members	abs_weight
Industrials	15	0.1931
Consumer Discretionary	11	0.1390
U.S. Mid Cap Equity	8	0.1157
U.S. Small Cap Growth	6	0.0850
U.S. Mid Cap Growth	6	0.0850
U.S. Small Cap Equity	5	0.0715
U.S. Momentum Equity	3	0.0378
Growth Equity	2	0.0274

U.S. Small Cap Momentum Equity	2	0.0269
Momentum Equity	2	0.0263
Unknown	1	0.0145
U.S. Extended Market Equity	1	0.0144
U.S. Small Cap ESG Equity	1	0.0144
U.S. Small Cap Multifactor Equity	1	0.0144
U.S. ESG Equity	1	0.0143
U.S. Size Factor Equity	1	0.0142
U.S. Mid Cap Quality Equity	1	0.0140
U.S. Mid Cap Momentum Equity	1	0.0136
Technology	1	0.0134
U.S. Small Cap Industrials	1	0.0133
U.S. Micro Cap Equity	1	0.0133
U.S. Moat Equity	1	0.0130
IPO Equity	1	0.0128
U.S. Small Cap Consumer Discretionary	1	0.0126

Replication basket – IJH (cor=0.98) basket lead at lag 1: -0.007

ticker	rep_wt	name	category	etf_signal
IJH	1.004	iShares Core S&P Mid-Cap ETF	U.S. Mid Cap Equity	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
28	SCHM	0.0146	Schwab U.S. Mid Cap ETF	U.S. Mid Cap Equity	LONG
50	VB	0.0146	Vanguard Small-Cap ETF	U.S. Small Cap Equity	FLAT
22	JHMM	0.0145	John Hancock Multifactor Mid Cap ETF	U.S. Mid Cap Equity	FLAT
23	IWR	0.0145	iShares Russell Mid-Cap ETF	U.S. Mid Cap Equity	LONG
35	MDY	0.0145	State Street SPDR S&P MIDCAP 400 ETF Tr...	U.S. Mid Cap Equity	FLAT
37	IJK	0.0145	iShares S&P Mid-Cap 400 Growth ETF	U.S. Mid Cap Growth	LONG
40	MDYG	0.0145	State Street SPDR S&P 400 Mid Cap Growt...	U.S. Mid Cap Growth	LONG
43	IVOO	0.0145	Vanguard S&P Mid-Cap 400 ETF		FLAT
48	IJH	0.0145	iShares Core S&P Mid-Cap ETF	U.S. Mid Cap Equity	FLAT
49	SPMD	0.0145	State Street SPDR Portfolio S&P 400 Mid...	U.S. Mid Cap Equity	FLAT
27	IMCB	0.0144	iShares Morningstar Mid-Cap ETF	U.S. Mid Cap Equity	LONG
32	NUSC	0.0144	Nuveen ESG Small-Cap ETF	U.S. Small Cap ESG Equity	FLAT
33	SMLF	0.0144	iShares U.S. Small-Cap Equity Factor ETF	U.S. Small Cap Multifact...	FLAT
39	IVOG	0.0144	Vanguard S&P Mid-Cap 400 Growth ETF	U.S. Mid Cap Growth	LONG
47	VXF	0.0144	Vanguard Extended Market ETF	U.S. Extended Market Equ...	FLAT
51	SCHA	0.0144	Schwab U.S. Small-Cap ETF	U.S. Small Cap Equity	LONG

5	IWM	0.0143	iShares Russell 2000 Index Fund	U.S. Small Cap Equity	FLAT
29	VO	0.0143	Vanguard Mid-Cap ETF	U.S. Mid Cap Equity	LONG
44	VTWO	0.0143	Vanguard Russell 2000 ETF	U.S. Small Cap Equity	FLAT
65	ETHO	0.0143	Amplify Etho Climate Leadership U.S. ETF	U.S. ESG Equity	FLAT
24	SIZE	0.0142	iShares MSCI USA Size Factor ETF	U.S. Size Factor Equity	LONG
36	IJT	0.0142	iShares S&P SmallCap 600 Growth ETF	U.S. Small Cap Growth	FLAT
38	VIOG	0.0142	Vanguard S&P Small-Cap 600 Growth ETF	U.S. Small Cap Growth	FLAT
41	SLYG	0.0142	State Street SPDR S&P 600 Small Cap Gro...	U.S. Small Cap Growth	FLAT
45	VBK	0.0142	Vanguard Small-Cap Growth ETF	U.S. Small Cap Growth	FLAT
46	ISCG	0.0142	iShares Morningstar Small-Cap Growth ETF	Growth Equity	FLAT
18	IMCG	0.0141	iShares Morningstar Mid-Cap Growth ETF	U.S. Mid Cap Growth	LONG
19	FXR	0.0141	First Trust Industrials AlphaDEX	Industrials	LONG
34	IWO	0.0141	iShares Russell 2000 Growth Fund	U.S. Small Cap Growth	LONG
42	VTWG	0.0141	Vanguard Russell 2000 Growth ETF	U.S. Small Cap Growth	LONG
60	PSC	0.0140	Principal U.S. Small-Cap ETF	U.S. Small Cap Equity	FLAT
68	XMHQ	0.0140	Invesco S&P MidCap Quality ETF	U.S. Mid Cap Quality Equ...	FLAT
21	IYJ	0.0139	iShares U.S. Industrials ETF	Industrials	FLAT
25	VIS	0.0139	Vanguard Industrials ETF	Industrials	FLAT
26	FIDU	0.0139	Fidelity MSCI Industrials Index ETF	Industrials	FLAT
13	IWP	0.0138	iShares Russell Midcap Growth ETF	U.S. Mid Cap Growth	LONG
17	VOT	0.0138	Vanguard Mid-Cap Growth ETF	U.S. Mid Cap Growth	LONG
67	PAVE	0.0137	Global X U.S. Infrastructure Developmen...	Industrials	FLAT
12	XMMO	0.0136	Invesco S&P MidCap Momentum ETF	U.S. Mid Cap Momentum Eq...	FLAT
52	FXD	0.0136	First Trust Cons. Discret. AlphaDEX	Consumer Discretionary	LONG
69	VFMO	0.0136	Vanguard U.S. Momentum Factor ETF	U.S. Momentum Equity	LONG
74	XSMO	0.0136	Invesco S&P SmallCap Momentum ETF	U.S. Small Cap Momentum ...	FLAT
59	RSPD	0.0135	Invesco S&P 500 Equal Weight Consumer D...	Consumer Discretionary	LONG
2	XLI	0.0134	State Street Industrial Select Sector S...	Industrials	FLAT
7	FTC	0.0134	First Trust Large Cap Growth AlphaDEX F...	Technology	LONG
8	PDP	0.0133	Invesco Dorsey Wright Momentum ETF	Momentum Equity	LONG
30	DWAS	0.0133	Invesco Dorsey Wright SmallCap Momentum...	U.S. Small Cap Momentum ...	FLAT
31	IWC	0.0133	iShares Microcap ETF	U.S. Micro Cap Equity	FLAT
66	PSCI	0.0133	Invesco S&P SmallCap Industrials ETF	U.S. Small Cap Industria...	FLAT
10	IYC	0.0132	iShares U.S. Consumer Discretionary ETF	Consumer Discretionary	LONG
11	RPG	0.0132	Invesco S&P 500 Pure Growth ETF	Growth Equity	LONG
54	FTXR	0.0132	First Trust Nasdaq Transportation ETF	Industrials	LONG
15	VCR	0.0131	Vanguard Consumer Discretion ETF	Consumer Discretionary	LONG
16	FDIS	0.0131	Fidelity MSCI Consumer Discretionary In...	Consumer Discretionary	LONG
70	PKB	0.0131	Invesco Building & Construction ETF	Industrials	LONG
6	FV	0.0130	First Trust Dorsey Wright Focus 5 ETF	Momentum Equity	LONG
20	MOAT	0.0130	VanEck Morningstar Wide Moat ETF	U.S. Moat Equity	FLAT
9	FPX	0.0128	First Trust US Equity Opportunities ETF	IPO Equity	FLAT
72	PRN	0.0128	Invesco Dorsey Wright Industrials Momen...	Industrials	FLAT
63	XTN	0.0127	State Street SPDR S&P Transportation ETF	Industrials	FLAT
1	XLY	0.0126	State Street Consumer Discretionary Sel...	Consumer Discretionary	LONG

53	PEJ	0.0126	Invesco Leisure and Entertainment ETF	Consumer Discretionary	LONG
73	PSCD	0.0126	Invesco S&P SmallCap Consumer Discretio...	U.S. Small Cap Consumer ...	LONG
3	XRT	0.0125	State Street SPDR S&P Retail ETF	Consumer Discretionary	LONG
58	IYT	0.0125	iShares U.S. Transportation ETF	Industrials	FLAT
62	XHB	0.0122	State Street SPDR S&P Homebuilders ETF	Consumer Discretionary	LONG
64	QMOM	0.0122	Alpha Architect U.S. Quantitative Momen...	U.S. Momentum Equity	LONG
14	MTUM	0.0120	iShares MSCI USA Momentum Factor ETF	U.S. Momentum Equity	LONG
61	XAR	0.0119	State Street SPDR S&P Aerospace & Defen...	Industrials	FLAT
55	PPA	0.0117	Invesco Aerospace & Defense ETF	Industrials	LONG
71	RTH	0.0116	VanEck Retail ETF	Consumer Discretionary	FLAT
4	JETS	0.0113	U.S. Global Jets ETF	Industrials	FLAT
56	ITA	0.0110	iShares U.S. Aerospace & Defense ETF	Industrials	LONG
57	ITB	0.0110	iShares U.S. Home Construction ETF	Consumer Discretionary	LONG

Cluster 2 – U.S. Large Cap Value

Members: 73 **Var explained:** 0.871 **Signal:** LONG since 2026-04-13 (57 days, 5.9%)

Signal logic:

- Latest buy_cross: 2026-04-13 – filter **PASS** (252d at date = +22.3%)
- Latest sell_cross: 2026-03-06
- Current 252d return: +18.1% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-04-13 > last sell at 2026-03-06)



Category breakdown

category	n_members	abs_weight
U.S. Dividend Equity	22	0.3012
U.S. Large Cap Value	10	0.1440
Value Equity	4	0.0558
U.S. Low Volatility Equity	4	0.0518
U.S. Multifactor Equity	3	0.0422
U.S. Value Factor Equity	3	0.0411
Dividend Equity	3	0.0401
Utilities	3	0.0382
U.S. Fundamental Equity	2	0.0287
U.S. Mid Cap Value	2	0.0286
U.S. Equity	2	0.0285
U.S. Equal Weight Equity	2	0.0284
U.S. Small Cap Low Volatility	2	0.0261
U.S. Mega Cap Value	1	0.0142
U.S. Revenue Weighted Equity	1	0.0142
Large Cap Blend Equities	1	0.0140
U.S. Buyback Equity	1	0.0138
U.S. Large Cap Equity	1	0.0137
U.S. Mid Cap Low Volatility	1	0.0135
U.S. Quality / Factor Equity	1	0.0135
U.S. Defensive Equity	1	0.0132
Global Equity	1	0.0127

Communication Services	1	0.0114
Real Estate	1	0.0111

Replication basket – VTV (cor=0.98) basket lead at lag 1: 0.005

ticker	rep_wt	name	category	etf_signal
VTV	0.993	Vanguard Value ETF	U.S. Large Cap Value	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
29	IWD	0.0145	iShares Russell 1000 Value ETF	U.S. Large Cap Value	LONG
31	JPUS	0.0145	JPMorgan Diversified Return U.S. Equity...	U.S. Multifactor Equity	FLAT
35	VONV	0.0145	Vanguard Russell 1000 Value ETF	U.S. Large Cap Value	LONG
3	RSP	0.0144	Invesco S&P 500 Equal Weight ETF	U.S. Equal Weight Equity	LONG
14	PRF	0.0144	Invesco RAFI US 1000 ETF	U.S. Fundamental Equity	LONG
21	NULV	0.0144	Nuveen ESG Large-Cap Value ETF	U.S. Large Cap Value	FLAT
30	IVE	0.0144	iShares S&P 500 Value ETF	U.S. Large Cap Value	FLAT
39	ILCV	0.0144	iShares Morningstar Large-Cap Value ETF	U.S. Large Cap Value	FLAT
40	IUSV	0.0144	iShares Core S&P U.S. Value ETF	U.S. Large Cap Value	FLAT
41	SCHV	0.0144	Schwab U.S. Large-Cap Value ETF	U.S. Large Cap Value	LONG
43	VTV	0.0144	Vanguard Value ETF	U.S. Large Cap Value	LONG
59	VOE	0.0144	Vanguard Mid-Cap Value ETF	U.S. Mid Cap Value	FLAT
64	DTD	0.0144	WisdomTree U.S. Total Dividend Fund	U.S. Dividend Equity	LONG
69	DGRO	0.0144	iShares Core Dividend Growth ETF	U.S. Dividend Equity	LONG
19	CFA	0.0143	VictoryShares US 500 Volatility Wtd ETF	U.S. Multifactor Equity	FLAT
25	FNDX	0.0143	Schwab Fundamental U.S. Large Company E...	U.S. Fundamental Equity	LONG
28	ONEV	0.0143	State Street SPDR Russell 1000 Low Vola...	U.S. Equity	FLAT
33	VOOV	0.0143	Vanguard S&P 500 Value ETF	U.S. Large Cap Value	FLAT
38	VYM	0.0143	Vanguard High Dividend Yield ETF	U.S. Dividend Equity	FLAT
42	SPYV	0.0143	State Street SPDR Portfolio S&P 500 Val...	U.S. Large Cap Value	FLAT
63	DLN	0.0143	WisdomTree U.S. LargeCap Dividend Fund	U.S. Dividend Equity	LONG
15	RWL	0.0142	Invesco S&P 500 Revenue ETF	U.S. Revenue Weighted Eq...	LONG
17	AIVL	0.0142	WisdomTree U.S. AI Enhanced Value Fund	U.S. Value Factor Equity	LONG
27	IWX	0.0142	iShares Russell Top 200 Value ETF	Value Equity	LONG
36	MGV	0.0142	Vanguard Mega Cap Value ETF	U.S. Mega Cap Value	LONG
44	FTA	0.0142	First Trust Large Cap Value AlphaDEX Fu...	Value Equity	LONG
55	IWS	0.0142	iShares Russell Mid-Cap Value ETF	U.S. Mid Cap Value	FLAT
56	ONEY	0.0142	State Street SPDR Russell 1000 Yield Fo...	U.S. Dividend Equity	FLAT
72	JPME	0.0142	JPMorgan Diversified Return U.S. Mid Ca...	U.S. Equity	FLAT
26	EQAL	0.0141	Invesco Russell 1000 Equal Weight ETF	U.S. Equal Weight Equity	FLAT

61	PFM	0.0141	Invesco Dividend Achievers ETF	U.S. Dividend Equity	FLAT
6	FEX	0.0140	First Trust Large Cap Core AlphaDEX Fund	Large Cap Blend Equities	LONG
22	SDY	0.0140	State Street SPDR S&P Dividend ETF	U.S. Dividend Equity	LONG
37	SCHD	0.0140	Schwab US Dividend Equity ETF	U.S. Dividend Equity	LONG
4	FVD	0.0139	First Trust VL Dividend	U.S. Dividend Equity	LONG
23	FDVV	0.0139	Fidelity High Dividend ETF	U.S. Dividend Equity	FLAT
47	SDOG	0.0139	ALPS Sector Dividend Dogs ETF	U.S. Dividend Equity	FLAT
49	DVY	0.0139	iShares Select Dividend ETF	U.S. Dividend Equity	FLAT
70	VIG	0.0139	Vanguard Div Appreciation ETF	U.S. Dividend Equity	LONG
5	PKW	0.0138	Invesco BuyBack Achievers ETF	U.S. Buyback Equity	LONG
10	PWV	0.0138	Invesco Large Cap Value ETF	Value Equity	LONG
73	JVAL	0.0138	JPMorgan U.S. Value Factor ETF	U.S. Value Factor Equity	LONG
66	DIA	0.0137	State Street SPDR Dow Jones Industrial ...	U.S. Large Cap Equity	LONG
11	RDVY	0.0136	First Trust Rising Dividend Achievers E...	U.S. Dividend Equity	FLAT
13	REGL	0.0136	ProShares S&P MidCap 400 Dividend Arist...	Dividend Equity	LONG
58	SPYD	0.0136	State Street SPDR Portfolio S&P 500 Hig...	U.S. Dividend Equity	FLAT
68	LGLV	0.0136	State Street SPDR US Large Cap Low Vola...	U.S. Low Volatility Equi...	LONG
16	DHS	0.0135	WisdomTree U.S. High Dividend Fund	U.S. Dividend Equity	FLAT
51	RPV	0.0135	Invesco S&P 500 Pure Value ETF	Value Equity	FLAT
53	XMLV	0.0135	Invesco S&P MidCap Low Volatility ETF	U.S. Mid Cap Low Volatil...	LONG
60	FTCS	0.0135	First Trust Capital Strength ETF	U.S. Quality / Factor Eq...	LONG
62	OUSA	0.0135	ALPS O'Shares U.S. Quality Dividend ETF	Dividend Equity	LONG
20	CFO	0.0134	VictoryShares US 500 Enhanced Volatilit...	U.S. Multifactor Equity	FLAT
57	SMMV	0.0134	iShares MSCI USA Small-Cap Min Vol Fact...	U.S. Small Cap Low Volat...	LONG
18	CDC	0.0132	VictoryShares US EQ Income Enhanced Vol...	U.S. Defensive Equity	FLAT
50	RDIV	0.0132	Invesco S&P Ultra Dividend Revenue ETF	U.S. Dividend Equity	FLAT
67	USMV	0.0132	iShares MSCI USA Min Vol Factor ETF	U.S. Low Volatility Equi...	LONG
32	VLUE	0.0131	iShares MSCI USA Value Factor ETF	U.S. Value Factor Equity	LONG
52	SPHD	0.0131	Invesco S&P 500 High Dividend Low Volat...	Dividend Equity	LONG
9	FIW	0.0130	First Trust Water ETF	Utilities	LONG
12	FDL	0.0130	First Trust Morningstar ETF	U.S. Dividend Equity	FLAT
45	PEY	0.0130	Invesco High Yield Equity Dividend Achi...	U.S. Dividend Equity	FLAT
7	PHO	0.0129	Invesco Water Resources ETF	Utilities	LONG
46	DIV	0.0129	Global X Super Dividend ETF	U.S. Dividend Equity	LONG
2	ACWV	0.0127	iShares MSCI Global Min Vol Factor ETF	Global Equity	LONG
24	LVHD	0.0127	Franklin U.S. Low Volatility High Divid...	U.S. Low Volatility Equi...	LONG
54	XSLV	0.0127	Invesco S&P SmallCap Low Volatility ETF	U.S. Small Cap Low Volat...	FLAT
34	HDV	0.0126	iShares Core High Dividend ETF	U.S. Dividend Equity	LONG
48	SMDV	0.0126	ProShares Russell 2000 Dividend Growers...	U.S. Dividend Equity	FLAT
8	CGW	0.0123	Invesco S&P Global Water Index ETF	Utilities	LONG
65	SPLV	0.0122	Invesco S&P 500 Low Volatility ETF	U.S. Low Volatility Equi...	LONG
71	IYZ	0.0114	iShares U.S. Telecommunications ETF	Communication Services	FLAT
1	REM	0.0111	iShares Mortgage Real Estate ETF	Real Estate	FLAT

Cluster 3 – U.S. Large Cap Growth**Members:** 67 **Var explained:** 0.816 **Signal:** FLAT**Signal logic:**

- Latest buy_cross: 2026-04-13 – filter **PASS** (252d at date = +34.1%)
- Latest sell_cross: 2026-06-26
- Current 252d return: +28.4% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-06-26 more recent than last effective buy 2026-04-13)

**Category breakdown**

category	n_members	abs_weight
Technology	34	0.5064
U.S. Large Cap Growth	11	0.1769
Communication Services	3	0.0421
U.S. Equity	2	0.0317
Growth Equity	2	0.0316
Convertible Bond	2	0.0291
Financials	2	0.0280
Consumer Discretionary	2	0.0272
U.S. Mega Cap Growth	1	0.0160
U.S. Equal Weight Equity	1	0.0159
IPO Equity	1	0.0145

Options Income	1	0.0145
U.S. Small Cap Technology	1	0.0145
Trend-Following Equity	1	0.0141
International IPO Equity	1	0.0130
U.S. Trend-Following Equity	1	0.0124
Israel Equity	1	0.0121

Replication basket – QQQ (cor=0.97) basket lead at lag 1: -0.016

ticker	rep_wt	name	category	etf_signal
QQQ	1.007	Invesco QQQ Trust, Series 1	U.S. Large Cap Growth	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
43	ONEQ	0.0163	Fidelity Nasdaq Composite Index ETF	U.S. Equity	LONG
16	QQQ	0.0162	Invesco QQQ Trust, Series 1	U.S. Large Cap Growth	FLAT
38	IGM	0.0162	iShares Expanded Tech Sector ETF	Technology	FLAT
55	ILCG	0.0162	iShares Morningstar Large-Cap Growth ETF	U.S. Large Cap Growth	LONG
31	XNTK	0.0161	State Street SPDR NYSE Technology ETF	Technology	LONG
40	IYW	0.0161	iShares U.S. Technology ETF	Technology	FLAT
42	NULG	0.0161	Nuveen ESG Large-Cap Growth ETF	U.S. Large Cap Growth	FLAT
46	IWF	0.0161	iShares Russell 1000 Growth Fund	U.S. Large Cap Growth	FLAT
48	VGT	0.0161	Vanguard Information Tech ETF	Technology	FLAT
52	FTEC	0.0161	Fidelity MSCI Information Technology In...	Technology	FLAT
53	VONG	0.0161	Vanguard Russell 1000 Growth ETF	U.S. Large Cap Growth	FLAT
57	SCHG	0.0161	Schwab U.S. Large-Cap Growth ETF	U.S. Large Cap Growth	LONG
59	VUG	0.0161	Vanguard Growth ETF	U.S. Large Cap Growth	FLAT
20	QTEC	0.0160	First Trust NASDAQ-100-Technology Secto...	Technology	LONG
26	FXL	0.0160	First Trust Technology AlphaDEX	Technology	FLAT
47	IVW	0.0160	iShares S&P 500 Growth ETF	U.S. Large Cap Growth	LONG
49	VOOG	0.0160	Vanguard S&P 500 Growth ETF	U.S. Large Cap Growth	LONG
54	MGK	0.0160	Vanguard Mega Cap Growth ETF	U.S. Mega Cap Growth	LONG
56	IUSG	0.0160	iShares Core S&P U.S. Growth ETF	U.S. Large Cap Growth	LONG
58	SPYG	0.0160	State Street SPDR Portfolio S&P 500 Gro...	U.S. Large Cap Growth	LONG
9	AIQ	0.0159	Global X Artificial Intelligence & Tech...	Technology	FLAT
19	QQEW	0.0159	First Trust Nasdaq-100 Select Equal Wei...	U.S. Equal Weight Equity	LONG
45	IWY	0.0159	iShares Russell Top 200 Growth ETF	Growth Equity	FLAT
1	XLK	0.0158	State Street Technology Select Sector S...	Technology	FLAT
23	XT	0.0158	iShares Future Exponential Technologies...	Technology	FLAT
39	IXN	0.0158	iShares Global Tech ETF	Technology	FLAT

37	PWB	0.0157	Invesco Large Cap Growth ETF	Growth Equity	FLAT
60	SPHB	0.0155	Invesco S&P 500 High Beta ETF	U.S. Equity	LONG
8	ROBO	0.0152	ROBO Global Robotics and Automation Ind...	Technology	LONG
22	TDIV	0.0152	First Trust NASDAQ Technology Dividend ...	Technology	FLAT
7	BOTZ	0.0151	Global X Robotics & Artificial Intellig...	Technology	FLAT
30	FDN	0.0151	First Trust DJ Internet Index Fund	Technology	FLAT
67	QTUM	0.0151	Defiance Quantum ETF	Technology	LONG
29	PNQI	0.0149	Invesco Nasdaq Internet ETF	Technology	FLAT
4	SMH	0.0148	VanEck Semiconductor ETF	Technology	LONG
6	SKYY	0.0148	First Trust Cloud Computing ETF	Technology	LONG
14	ARKQ	0.0147	ARK Autonomous Technology & Robotics ETF	Technology	LONG
41	CWB	0.0147	State Street SPDR Bloomberg Convertible...	Convertible Bond	FLAT
3	SOXX	0.0146	iShares PHLX SOX Semiconductor Sector I...	Technology	LONG
17	NXTG	0.0146	First Trust Indxx NextG ETF	Technology	FLAT
24	XSD	0.0146	State Street SPDR S&P Semiconductor ETF	Technology	LONG
28	IPO	0.0145	Renaissance IPO ETF	IPO Equity	LONG
36	QYLD	0.0145	Global X NASDAQ 100 Covered Call ETF	Options Income	FLAT
63	PSCT	0.0145	Invesco S&P SmallCap Information Techno...	U.S. Small Cap Technology	LONG
66	PTF	0.0145	Invesco Dorsey Wright Technology Moment...	Technology	LONG
5	IGV	0.0144	iShares Expanded Tech-Software Sector E...	Technology	FLAT
13	ARKW	0.0144	ARK Next Generation Internet ETF	Technology	LONG
34	FINX	0.0144	Global X FinTech ETF	Financials	FLAT
44	ICVT	0.0144	iShares Convertible Bond ETF	Convertible Bond	FLAT
21	PSI	0.0143	Invesco Semiconductors ETF	Technology	LONG
65	FTXL	0.0143	First Trust Nasdaq Semiconductor ETF	Technology	LONG
12	ARKK	0.0142	ARK Innovation ETF	Technology	LONG
50	VOX	0.0142	Vanguard Communication Services ETF	Communication Services	FLAT
51	FCOM	0.0142	Fidelity MSCI Communication Services In...	Communication Services	FLAT
35	PTNQ	0.0141	Pacer Trendpilot 100 ETF	Trend-Following Equity	FLAT
25	IBUY	0.0139	Amplify Online Retail ETF	Consumer Discretionary	LONG
27	CIBR	0.0138	First Trust NASDAQ Cybersecurity ETF	Technology	FLAT
2	XLC	0.0137	State Street Communication Services Sel...	Communication Services	FLAT
18	PRNT	0.0137	3D Printing (The) ETF	Technology	LONG
11	HACK	0.0136	Amplify Cybersecurity ETF	Technology	FLAT
32	IPAY	0.0136	Amplify Digital Payments ETF	Financials	FLAT
15	ESPO	0.0133	VanEck Video Gaming and eSports ETF	Consumer Discretionary	FLAT
10	BLOK	0.0131	Amplify Blockchain Technology ETF	Technology	FLAT
33	FPXI	0.0130	First Trust International Equity Opport...	International IPO Equity	FLAT
61	PTLC	0.0124	Pacer Trendpilot US Large Cap ETF	U.S. Trend-Following Equ...	FLAT
62	EIS	0.0121	iShares MSCI Israel ETF	Israel Equity	FLAT
64	BLCN	0.0115	Siren NexGen Economy ETF	Technology	FLAT

Cluster 4 – Developed ex-US Equity

Members: 60 **Var explained:** 0.898 **Signal:** LONG since 2026-07-06 (0 days, 0.0%)

Signal logic:

- Latest buy_cross: 2026-07-06 – filter **PASS** (252d at date = +21.5%)
- Latest sell_cross: 2026-06-25
- Current 252d return: +21.5% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-07-06 > last sell at 2026-06-25)

**Category breakdown**

category	n_members	abs_weight
International Equity	19	0.3205
Developed ex-US Equity	9	0.1551
Global ex-US Equity	5	0.0861
Europe Equity	4	0.0684
International Dividend Equity	4	0.0669
Alternatives	4	0.0625
Eurozone Equity	3	0.0497
Commodity	1	0.0174
Global ex-US Small Cap Equity	1	0.0168
Germany Equity	1	0.0164
France Equity	1	0.0164

Italy Equity	1	0.0161
United Kingdom Equity	1	0.0160
Europe Financials	1	0.0160
Netherlands Equity	1	0.0159
Sweden Equity	1	0.0158
Spain Equity	1	0.0153
Austria Equity	1	0.0150
Poland Equity	1	0.0137

Replication basket – VEA (cor=0.99) basket lead at lag 1: -0.018

ticker	rep_wt	name	category	etf_signal
VEA	0.964	Vanguard FTSE Developed Markets ETF	Developed ex-US Equity	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
2	EFA	0.0175	iShares MSCI EAFE ETF	Developed ex-US Equity	LONG
3	VEA	0.0175	Vanguard FTSE Developed Markets ETF	Developed ex-US Equity	FLAT
41	ESGD	0.0175	iShares ESG Aware MSCI EAFE ETF	International Equity	LONG
51	IEFA	0.0175	iShares Core MSCI EAFE ETF	Developed ex-US Equity	LONG
40	GSIE	0.0174	Goldman Sachs ActiveBeta International ...	Commodity	LONG
52	SCHF	0.0174	Schwab International Equity ETF	Developed ex-US Equity	FLAT
53	SPDW	0.0174	State Street SPDR Portfolio Developed W...	Developed ex-US Equity	LONG
16	DWM	0.0173	WisdomTree International Equity Fund	International Dividend E...	FLAT
25	TLTD	0.0173	FlexShares Morningstar Developed Market...	International Equity	FLAT
32	INTF	0.0173	iShares International Equity Factor ETF	International Equity	FLAT
47	IXUS	0.0173	iShares Core MSCI Total International S...	Global ex-US Equity	LONG
49	VGK	0.0173	Vanguard FTSEEuropean ETF	Europe Equity	LONG
6	IEV	0.0172	iShares Europe ETF	Europe Equity	LONG
19	PXF	0.0172	Invesco RAFI Developed Markets ex-U.S. ...	International Equity	FLAT
29	ACWX	0.0172	iShares MSCI ACWI ex U.S. ETF	Global ex-US Equity	FLAT
30	CWI	0.0172	State Street SPDR MSCI ACWI ex-US ETF	Global ex-US Equity	FLAT
33	IQLT	0.0172	iShares MSCI Intl Quality Factor ETF	International Equity	LONG
39	FNDF	0.0172	Schwab Fundamental International Equity...	International Equity	FLAT
46	IEUR	0.0172	iShares Core MSCI Europe ETF	Europe Equity	LONG
48	VEU	0.0172	Vanguard FTSE All World Ex US ETF	Global ex-US Equity	FLAT
50	VXUS	0.0172	Vanguard Total International Stock ETF	Global ex-US Equity	FLAT
35	QEFA	0.0171	State Street SPDR MSCI EAFE StrategicFa...	International Equity	LONG
14	EZU	0.0170	iShares MSCI Eurozone ETF	Eurozone Equity	FLAT
23	EFV	0.0170	iShares MSCI EAFE VaValue ETF	Developed ex-US Equity	FLAT

26	JPIN	0.0170	JPMorgan Diversified Return Internation...	International Equity	FLAT
38	VYMI	0.0170	Vanguard International High Dividend Yi...	International Dividend E...	FLAT
44	SCHC	0.0170	Schwab International Small-Cap Equity E...	Developed ex-US Equity	FLAT
18	IQDF	0.0169	FlexShares International Quality Divide...	International Equity	LONG
22	SCZ	0.0169	iShares MSCI EAFE Small-Cap ETF	Developed ex-US Equity	FLAT
24	FNDC	0.0169	Schwab Fundamental International Small ...	International Equity	FLAT
34	IVLU	0.0169	iShares MSCI Intl Value Factor ETF	International Equity	FLAT
37	RODM	0.0169	Hartford Multifactor Developed Markets ...	International Equity	FLAT
56	HFXI	0.0169	NYLI FTSE International Equity Currency...	International Equity	FLAT
20	EFG	0.0168	iShares MSCI EAFE Growth ETF	Developed ex-US Equity	FLAT
36	FEZ	0.0168	State Street SPDR EURO STOXX 50 ETF	Eurozone Equity	FLAT
45	VSS	0.0168	Vanguard FTSE All-Wld ex-US SmCp Idx ETF	Global ex-US Small Cap E...	FLAT
4	FEP	0.0167	First Trust Europe AlphaDEX Fund	Europe Equity	FLAT
7	DLS	0.0167	WisdomTree International SmallCap Fund	International Equity	FLAT
31	IMTM	0.0166	iShares MSCI Intl Momentum Factor ETF	International Equity	FLAT
43	VIGI	0.0166	Vanguard International Dividend Appreci...	International Dividend E...	LONG
1	EWG	0.0164	iShares MSCI Germany Index Fund	Germany Equity	FLAT
12	EWQ	0.0164	iShares MSCI France Index Fund	France Equity	LONG
21	GWX	0.0164	State Street SPDR S&P International Sma...	International Equity	FLAT
5	VIDI	0.0162	Vident International Equity Strategy ETF	International Equity	FLAT
59	IPKW	0.0162	Invesco International BuyBack Achievers...	International Equity	FLAT
60	EWI	0.0161	iShares MSCI Italy ETF	Italy Equity	FLAT
13	EWU	0.0160	iShares MSCI United Kingdom ETF	United Kingdom Equity	FLAT
15	IDV	0.0160	iShares International Select Dividend E...	International Dividend E...	FLAT
17	EUFN	0.0160	iShares MSCI Europe Financials ETF	Europe Financials	LONG
10	HEZU	0.0159	iShares Currency Hedged MSCI Eurozone E...	Eurozone Equity	FLAT
27	DBEF	0.0159	Xtrackers MSCI EAFE Hedged Equity ETF	Alternatives	FLAT
42	HDEF	0.0159	Xtrackers MSCI EAFE High Dividend Yield...	International Equity	FLAT
58	EWN	0.0159	iShares MSCI Netherlands Index Fund	Netherlands Equity	LONG
28	HEFA	0.0158	iShares Currency Hedged MSCI EAFE ETF	Alternatives	FLAT
55	EWD	0.0158	iShares MSCI Sweden ETF	Sweden Equity	FLAT
8	HEDJ	0.0154	WisdomTree Europe Hedged Equity Fund	Alternatives	FLAT
9	IHDG	0.0153	WisdomTree International Hedged Quality...	Alternatives	LONG
11	EWP	0.0153	iShares MSCI Spain ETF	Spain Equity	FLAT
57	EWO	0.0150	iShares MSCI Austria ETF	Austria Equity	LONG
54	EPOL	0.0137	iShares MSCI Poland ETF	Poland Equity	FLAT

Cluster 5 – U.S. Large Cap Equity**Members:** 46 **Var explained:** 0.951 **Signal:** FLAT**Signal logic:**

- Latest buy_cross: 2026-04-13 – filter **PASS** (252d at date = +25.8%)
- Latest sell_cross: 2026-06-24
- Current 252d return: +20.2% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-06-24 more recent than last effective buy 2026-04-13)



Category breakdown

category	n_members	abs_weight
U.S. Large Cap Equity	9	0.1998
U.S. Total Market Equity	7	0.1555
Global Equity	7	0.1526
Asset Allocation	4	0.0808
U.S. Mega Cap Equity	3	0.0654
U.S. Quality / Factor Equity	3	0.0646
U.S. Multifactor Equity	3	0.0646
U.S. ESG Equity	2	0.0443
U.S. Equity	2	0.0442
Commodity	1	0.0222
Large Cap Value Equities	1	0.0220
Large Cap Blend Equities	1	0.0220
Dividend Equity	1	0.0217
U.S. Dividend Equity	1	0.0213
Alternatives	1	0.0189

Replication basket – VOO (cor=1.00) basket lead at lag 1: -0.010

ticker	rep_wt	name	category	etf_signal
VOO	0.935	Vanguard S&P 500 ETF	U.S. Large Cap Equity	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
1	SPY	0.0222	State Street SPDR S&P 500 ETF Trust	U.S. Large Cap Equity	LONG
20	IWV	0.0222	iShares Russell 3000 Fund	U.S. Total Market Equity	FLAT
21	IYY	0.0222	iShares Dow Jones U.S. ETF	U.S. Total Market Equity	FLAT
24	SPYX	0.0222	State Street SPDR S&P 500 Fossil Fuel R...	U.S. Equity	LONG
26	ESGU	0.0222	iShares ESG Aware MSCI USA ETF	U.S. ESG Equity	FLAT
27	IWB	0.0222	iShares Russell 1000 ETF	U.S. Large Cap Equity	FLAT
32	VTHR	0.0222	Vanguard Russell 3000 ETF	U.S. Total Market Equity	FLAT
33	GSLC	0.0222	Goldman Sachs ActiveBeta U.S. Large Cap...	Commodity	LONG
35	VONE	0.0222	Vanguard Russell 1000 ETF	U.S. Large Cap Equity	FLAT
37	VV	0.0222	Vanguard Large-Cap ETF	U.S. Large Cap Equity	FLAT
38	ILCB	0.0222	iShares Morningstar Large-Cap ETF	U.S. Large Cap Equity	FLAT
39	ITOT	0.0222	iShares Core S&P Total U.S. Stock Marke...	U.S. Total Market Equity	FLAT
40	IVV	0.0222	iShares Core S&P 500 ETF	U.S. Large Cap Equity	LONG
41	SCHB	0.0222	Schwab U.S. Broad Market ETF	U.S. Total Market Equity	FLAT
42	SCHX	0.0222	Schwab U.S. Large-Cap ETF	U.S. Large Cap Equity	FLAT
43	SPTM	0.0222	State Street SPDR Portfolio S&P 1500 Co...	U.S. Total Market Equity	FLAT
44	VOO	0.0222	Vanguard S&P 500 ETF	U.S. Large Cap Equity	LONG
45	VTI	0.0222	Vanguard Total Stock Market ETF	U.S. Total Market Equity	FLAT
8	URTH	0.0221	iShares MSCI World ETF	Global Equity	LONG
16	JHML	0.0221	John Hancock Multifactor Large Cap ETF	U.S. Large Cap Equity	FLAT
19	SUSA	0.0221	iShares ESG Optimized MSCI USA ETF	U.S. ESG Equity	FLAT
28	IWL	0.0221	iShares Russell Top 200 ETF	U.S. Equity	LONG
36	MGC	0.0221	Vanguard Mega Cap ETF	U.S. Mega Cap Equity	LONG
18	DSI	0.0220	iShares ESG MSCI KLD 400 ETF	Large Cap Blend Equities	FLAT
22	LRGF	0.0220	iShares U.S. Equity Factor ETF	U.S. Multifactor Equity	FLAT
34	EPS	0.0220	WisdomTree U.S. LargeCap Fund	Large Cap Value Equities	FLAT
3	ACWI	0.0219	iShares MSCI ACWI ETF	Global Equity	FLAT
9	CRBN	0.0219	iShares Low Carbon Optimized MSCI ACWI ...	Global Equity	LONG
11	VT	0.0219	Vanguard Total World Stock Index ETF	Global Equity	FLAT
12	TILT	0.0219	FlexShares Mornigstar US Market Factors...	U.S. Multifactor Equity	FLAT
10	SPGM	0.0218	State Street SPDR Portfolio MSCI Global...	Global Equity	FLAT
23	OEF	0.0218	iShares S&P 100 Fund	U.S. Mega Cap Equity	LONG
29	QUAL	0.0218	iShares MSCI USA Quality Factor ETF	U.S. Quality / Factor Eq...	FLAT
4	AOA	0.0217	iShares Core 80/20 Aggressive Allocatio...	Asset Allocation	FLAT

14	QDF	0.0217	FlexShares Quality Dividend Index Fund	Dividend Equity	FLAT
30	QUS	0.0216	State Street SPDR MSCI USA StrategicFac...	U.S. Quality / Factor Eq...	FLAT
13	I00	0.0215	iShares Global 100 ETF	Global Equity	FLAT
15	CATH	0.0215	Global X S&P 500 Catholic Values ETF	Global Equity	LONG
25	XLG	0.0214	Invesco S&P 500 Top 50 ETF	U.S. Mega Cap Equity	FLAT
17	DGRW	0.0213	WisdomTree U.S. Quality Dividend Growth...	U.S. Dividend Equity	LONG
31	SPHQ	0.0213	Invesco S&P 500 Quality ETF	U.S. Quality / Factor Eq...	LONG
7	AOR	0.0211	iShares Core 60/40 Balanced Allocation ...	Asset Allocation	FLAT
46	OMFL	0.0207	Invesco Russell 1000 Dynamic Multifacto...	U.S. Multifactor Equity	FLAT
6	AOM	0.0198	iShares Core 40/60 Moderate Allocation ...	Asset Allocation	FLAT
2	QAI	0.0189	NYLI Hedge Multi-Strategy Tracker ETF	Alternatives	FLAT
5	AOK	0.0182	iShares Core 30/70 Conservative Allocat...	Asset Allocation	FLAT

Cluster 6 – Aggregate Fixed Income

Members: 43 **Var explained:** 0.842 **Signal:** LONG since 2026-06-25 (6 days, -0.4%)

Signal logic:

- Latest buy_cross: 2026-06-25 – filter **PASS** (252d at date = +3.8%)
- Latest sell_cross: 2026-06-08
- Current 252d return: +3.1% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-06-25 > last sell at 2026-06-08)



Category breakdown

category	n_members	abs_weight
Fixed Income	13	0.3082
Aggregate Fixed Income	6	0.1478
Investment Grade Corporate Bond	6	0.1441
Mortgage-Backed Fixed Income	6	0.1326
Active Fixed Income	4	0.0943
Cash / Ultra-Short Fixed Income	3	0.0667
Target Maturity Bond	2	0.0420
International Fixed Income	2	0.0417
Agency Fixed Income	1	0.0227

Replication basket – BND (cor=0.99) basket lead at lag 1: 0.009

ticker	rep_wt	name	category	etf_signal
BND	0.729	Vanguard Total Bond Market ETF	Aggregate Fixed Income	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
13	BIV	0.0250	Vanguard Intermediate-Term Bond ETF	Fixed Income	LONG
16	AGG	0.0249	iShares Core U.S. Aggregate Bond ETF	Aggregate Fixed Income	LONG
33	VGIT	0.0249	Vanguard Intermediate-Term Treasury ETF	Fixed Income	LONG
17	BND	0.0248	Vanguard Total Bond Market ETF	Aggregate Fixed Income	LONG
19	SCHZ	0.0248	Schwab US Aggregate Bond ETF	Aggregate Fixed Income	LONG
29	IEI	0.0248	iShares 3-7 Year Treasury Bond ETF	Fixed Income	LONG
30	SPTI	0.0248	State Street SPDR Portfolio Intermediat...	Fixed Income	LONG
32	SCHR	0.0248	Schwab Intermediate-Term U.S. Treasury ...	Fixed Income	LONG
6	GVI	0.0247	iShares Intermediate Government/Credit ...	Fixed Income	LONG
12	IUSB	0.0247	iShares Core Universal USD Bond ETF	Aggregate Fixed Income	LONG
20	SPAB	0.0247	State Street SPDR Portfolio Aggregate B...	Aggregate Fixed Income	LONG
1	IEF	0.0246	iShares 7-10 Year Treasury Bond ETF	Fixed Income	LONG
11	ISTB	0.0245	iShares Core 1-5 Year USD Bond ETF	Fixed Income	LONG
9	SPIB	0.0243	State Street SPDR Portfolio Intermediat...	Investment Grade Corpora...	LONG
10	IGIB	0.0243	iShares 5-10 Year Investment Grade Corp...	Investment Grade Corpora...	LONG
14	BSV	0.0242	Vanguard Short-Term Bond ETF	Fixed Income	LONG
15	VCIT	0.0242	Vanguard Intermediate-Term Corporate Bo...	Investment Grade Corpora...	LONG
35	MBB	0.0242	iShares MBS ETF	Mortgage-Backed Fixed In...	LONG
36	VMBS	0.0242	Vanguard Mortgage-Backed Securities ETF	Mortgage-Backed Fixed In...	LONG
5	FBND	0.0241	Fidelity Total Bond ETF	Active Fixed Income	LONG
37	SPMB	0.0241	State Street SPDR Portfolio Mortgage Ba...	Mortgage-Backed Fixed In...	LONG
24	IGSB	0.0240	iShares 1-5 Year Investment Grade Corpo...	Investment Grade Corpora...	LONG
26	VCSH	0.0239	Vanguard Short-Term Corporate Bond ETF	Investment Grade Corpora...	LONG
3	BOND	0.0238	PIMCO Active Bond Exchange-Traded Fund ...	Active Fixed Income	LONG
4	GTO	0.0238	Invesco Total Return Bond ETF	Active Fixed Income	LONG
7	AGGY	0.0238	WisdomTree Yield Enhanced U.S. Aggregat...	Aggregate Fixed Income	LONG
25	SLQD	0.0234	iShares 0-5 Year Investment Grade Corpo...	Investment Grade Corpora...	LONG
31	GOVT	0.0233	iShares U.S. Treasury Bond ETF	Fixed Income	LONG
34	GNMA	0.0233	iShares GNMA Bond ETF	Mortgage-Backed Fixed In...	LONG
38	SHY	0.0229	iShares 1-3 Year Treasury Bond ETF	Fixed Income	FLAT
28	AGZ	0.0227	iShares Agency Bond ETF	Agency Fixed Income	LONG
18	TOTL	0.0226	State Street DoubleLine Total Return Ta...	Active Fixed Income	LONG
23	SPSB	0.0226	State Street SPDR Portfolio Short Term ...	Fixed Income	LONG
41	VGSH	0.0225	Vanguard Short-Term Treasury ETF	Cash / Ultra-Short Fixed...	FLAT
40	SCHO	0.0221	Schwab Short-Term U.S. Treasury ETF	Cash / Ultra-Short Fixed...	FLAT
39	SPTS	0.0220	State Street SPDR Portfolio Short Term ...	Cash / Ultra-Short Fixed...	FLAT
22	BSCQ	0.0213	Invesco BulletShares 2026 Corporate Bon...	Target Maturity Bond	FLAT
42	BNDX	0.0210	Vanguard Total International Bond ETF	International Fixed Inco...	LONG
8	IBDR	0.0207	iShares iBonds Dec 2026 Term Corporate ...	Target Maturity Bond	FLAT
43	IAGG	0.0207	iShares International Aggregate Bond Fu...	International Fixed Inco...	LONG
27	CMBS	0.0186	iShares CMBS Bond ETF	Mortgage-Backed Fixed In...	LONG
2	LMBS	0.0181	First Trust Low Duration Opportunities ...	Mortgage-Backed Fixed In...	FLAT
21	LDUR	0.0172	PIMCO Enhanced Low Duration Active Exch...	Fixed Income	LONG

Cluster 7 – U.S. Small Cap Equity**Members:** 42 **Var explained:** 0.888 **Signal:** FLAT**Signal logic:**

- Latest buy_cross: 2026-04-10 – filter **PASS** (252d at date = +37.1%)
- Latest sell_cross: 2026-05-14
- Current 252d return: +21.4% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-14 more recent than last effective buy 2026-04-10)

**Category breakdown**

category	n_members	abs_weight
Financials	13	0.2923
U.S. Small Cap Value	6	0.1480
U.S. Small Cap Equity	5	0.1230
U.S. Mid Cap Value	3	0.0745
Leveraged / Inverse	2	0.0479
U.S. Small Cap Fundamental Equity	1	0.0248
Mid Cap Value Equities	1	0.0248
Dividend Equity	1	0.0247
U.S. Mid Cap Dividend Equity	1	0.0246
Small Cap Value Equities	1	0.0246
U.S. Mid/Small Cap Equity	1	0.0246

U.S. Small/Mid Cap Fundamental Equity	1	0.0245
U.S. Small Cap Dividend Equity	1	0.0244
Value Equity	1	0.0244
U.S. Mid Cap Equity	1	0.0243
U.S. Small/Mid Cap Equity	1	0.0242
U.S. Equity	1	0.0238
Global Equity	1	0.0206

Replication basket – IJR (cor=0.98) basket lead at lag 1: -0.002

ticker	rep_wt	name	category	etf_signal
IJR	0.93	iShares Core S&P Small-Cap ETF	U.S. Small Cap Equity	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
31	VBR	0.0250	Vanguard Small-Cap Value ETF	U.S. Small Cap Value	FLAT
24	IVOV	0.0249	Vanguard S&P Mid-Cap 400 Value ETF	U.S. Mid Cap Value	FLAT
25	MDYV	0.0249	State Street SPDR S&P 400 Mid Cap Value...	U.S. Mid Cap Value	FLAT
18	EZM	0.0248	WisdomTree U.S. MidCap Fund	Mid Cap Value Equities	FLAT
20	FNDA	0.0248	Schwab Fundamental U.S. Small Company E...	U.S. Small Cap Fundament...	FLAT
22	IJJ	0.0248	iShares S&P Mid-Cap 400 Value ETF	U.S. Mid Cap Value	FLAT
8	VIOO	0.0247	Vanguard S&P Small-Cap 600 ETF	U.S. Small Cap Equity	FLAT
9	IJR	0.0247	iShares Core S&P Small-Cap ETF	U.S. Small Cap Equity	FLAT
10	SPSM	0.0247	State Street SPDR Portfolio S&P 600 Sma...	U.S. Small Cap Equity	FLAT
21	IWN	0.0247	iShares Russell 2000 Value ETF	U.S. Small Cap Value	FLAT
41	SDVY	0.0247	First Trust SMID Cap Rising Dividend Ac...	Dividend Equity	LONG
5	FNX	0.0246	First Trust Mid Cap Core AlphaDEX Fund	U.S. Mid/Small Cap Equity	FLAT
6	FYX	0.0246	First Trust Small Cap Core AlphaDEX Fund	U.S. Small Cap Equity	FLAT
16	DON	0.0246	WisdomTree U.S. MidCap Dividend Fund	U.S. Mid Cap Dividend Eq...	FLAT
17	EES	0.0246	WisdomTree U.S. SmallCap Fund	Small Cap Value Equities	FLAT
23	IJS	0.0246	iShares S&P SmallCap 600 Value ETF	U.S. Small Cap Value	LONG
26	SLYV	0.0246	State Street SPDR S&P 600 Small Cap Val...	U.S. Small Cap Value	FLAT
27	VIOV	0.0246	Vanguard S&P Small-Cap 600 Value ETF	U.S. Small Cap Value	FLAT
28	VTWV	0.0246	Vanguard Russell 2000 Value ETF	U.S. Small Cap Value	FLAT
7	PRFZ	0.0245	Invesco RAFI US 1500 Small-Mid ETF	U.S. Small/Mid Cap Funda...	FLAT
15	DES	0.0244	WisdomTree U.S. SmallCap Dividend Fund	U.S. Small Cap Dividend ...	FLAT
40	FYT	0.0244	First Trust Small Cap Value AlphaDEX Fu...	Value Equity	FLAT
32	RWJ	0.0243	Invesco S&P SmallCap 600 Revenue ETF	U.S. Small Cap Equity	FLAT
35	RWK	0.0243	Invesco S&P MidCap 400 Revenue ETF	U.S. Mid Cap Equity	FLAT
33	RFV	0.0242	Invesco S&P MidCap 400 Pure Value ETF	Leveraged / Inverse	FLAT

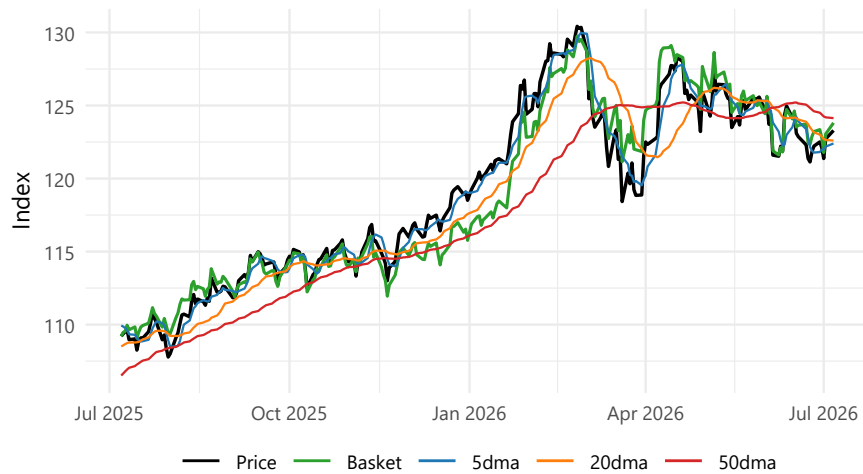
42	ISMD	0.0242	Inspire Small/Mid Cap ETF	U.S. Small/Mid Cap Equity	LONG
36	OMFS	0.0238	Invesco Russell 2000 Dynamic Multifacto...	U.S. Equity	FLAT
37	RZV	0.0238	Invesco S&P SmallCap 600 Pure Value ETF	Leveraged / Inverse	FLAT
11	FXO	0.0237	First Trust Financials AlphaDEX	Financials	LONG
29	VFH	0.0232	Vanguard Financials ETF	Financials	FLAT
30	FNCL	0.0232	Fidelity MSCI Financials Index ETF	Financials	LONG
3	KBE	0.0230	State Street SPDR S&P Bank ETF	Financials	FLAT
13	IYF	0.0230	iShares U.S. Financial ETF	Financials	FLAT
14	IYG	0.0229	iShares U.S. Financial Services ETF	Financials	FLAT
39	KCE	0.0229	State Street SPDR S&P Capital Markets E...	Financials	FLAT
19	KBWB	0.0225	Invesco KBW Bank ETF	Financials	FLAT
1	XLF	0.0224	State Street Financial Select Sector SP...	Financials	LONG
2	KRE	0.0223	State Street SPDR S&P Regional Banking ...	Financials	FLAT
12	IAT	0.0222	iShares U.S. Regional Banks ETF	Financials	LONG
38	IAI	0.0214	iShares U.S. Broker-Dealers & Securitie...	Financials	LONG
34	PSP	0.0206	Invesco Global Listed Private Equity ETF	Global Equity	FLAT
4	KIE	0.0195	State Street SPDR S&P Insurance ETF	Financials	LONG

Cluster 8 – Asia-Pacific Equity

Members: 25 **Var explained:** 0.680 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-10 – filter **PASS** (252d at date = +38.9%)
- Latest sell_cross: 2026-06-23
- Current 252d return: +11.7% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-06-23 more recent than last effective buy 2026-04-10)



Category breakdown

category	n_members	abs_weight
Asia-Pacific Equity	2	0.0905
International Equity	2	0.0886
International Dividend Equity	2	0.0885
Real Estate	2	0.0848
Global Equity	2	0.0848
Emerging Markets Fixed Income	2	0.0771
Currency	2	0.0744
Australia Equity	1	0.0448
Canada Equity	1	0.0425
Switzerland Equity	1	0.0400
Belgium Equity	1	0.0398
Singapore Equity	1	0.0384
South Africa Equity	1	0.0378
New Zealand Equity	1	0.0363
Latin America Equity	1	0.0363
Mexico Equity	1	0.0344
Brazil Equity	1	0.0306
Chile Equity	1	0.0305

Replication basket – EWZ+IGF+BBAX (cor=0.97) basket lead at lag 1: -0.002

ticker	rep_wt	name	category	etf_signal
EWZ	0.087	iShares MSCI Brazil ETF	Brazil Equity	FLAT
IGF	0.289	iShares Global Infrastructure ETF	Global Equity	LONG
BBAX	0.463	JPMorgan BetaBuilders Developed Asia ...	Asia-Pacific Equity	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
8	EPP	0.0453	iShares MSCI Pacific Ex-Japan Index Fund	Asia-Pacific Equity	FLAT
12	IDLV	0.0452	Invesco S&P International Developed Low...	International Equity	FLAT
25	BBAX	0.0452	JPMorgan BetaBuilders Developed Asia Pa...	Asia-Pacific Equity	FLAT
4	EWA	0.0448	iShares MSCI Australia Index Fund	Australia Equity	FLAT
11	DWX	0.0446	State Street SPDR S&P International Div...	International Dividend E...	FLAT
3	PID	0.0439	Invesco International Dividend Achiever...	International Dividend E...	FLAT
13	EFAV	0.0434	iShares MSCI EAFE Min Vol Factor ETF	International Equity	FLAT
9	NFRA	0.0430	FlexShares STOXX Global Broad Infrastru...	Global Equity	FLAT
14	VNQI	0.0429	Vanguard Global ex-U.S. Real Estate ETF	Real Estate	FLAT
7	EWC	0.0425	iShares MSCI Canada Index Fund	Canada Equity	FLAT
2	RWX	0.0419	State Street SPDR Dow Jones Internation...	Real Estate	FLAT
10	IGF	0.0418	iShares Global Infrastructure ETF	Global Equity	LONG
5	EWL	0.0400	iShares MSCI Switzerland ETF	Switzerland Equity	LONG
24	EWK	0.0398	iShares MSCI Belgium ETF	Belgium Equity	FLAT
20	FXA	0.0395	Invesco CurrencyShares Australian Dolla...	Currency	FLAT
15	EBND	0.0387	State Street SPDR Bloomberg Emerging Ma...	Emerging Markets Fixed I...	LONG
16	EMLC	0.0385	VanEck J. P. Morgan EM Local Currency B...	Emerging Markets Fixed I...	LONG
6	EWS	0.0384	iShares MSCI Singapore ETF	Singapore Equity	LONG
22	EZA	0.0378	iShares MSCI South Africa Index Fund	South Africa Equity	FLAT
19	ILF	0.0363	iShares Latin America 40 ETF	Latin America Equity	FLAT
21	ENZL	0.0363	iShares MSCI New Zealand ETF	New Zealand Equity	FLAT
23	FXC	0.0349	Invesco CurrencyShares Canadian Dollar ...	Currency	FLAT
17	EWX	0.0344	iShares MSCI Mexico ETF	Mexico Equity	FLAT
1	EWZ	0.0306	iShares MSCI Brazil ETF	Brazil Equity	FLAT
18	ECH	0.0305	iShares MSCI Chile ETF	Chile Equity	FLAT

Cluster 9 – Emerging Markets Equity

Members: 25 **Var explained:** 0.872 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-10 – filter **PASS** (252d at date = +57.0%)
- Latest sell_cross: 2026-07-02
- Current 252d return: +33.1% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-07-02 more recent than last effective buy 2026-04-10)



Category breakdown

category	n_members	abs_weight
Emerging Markets Equity	19	0.7727
Asia Equity	2	0.0814
Asia ex-Japan Equity	1	0.0418
Taiwan Equity	1	0.0361
South Korea Equity	1	0.0341
Global Dividend Equity	1	0.0338

Replication basket – IEMG (cor=0.99) basket lead at lag 1: -0.030

ticker	rep_wt	name	category	etf_signal
IEMG	0.921	iShares Core MSCI Emerging Markets ETF	Emerging Markets Equi...	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

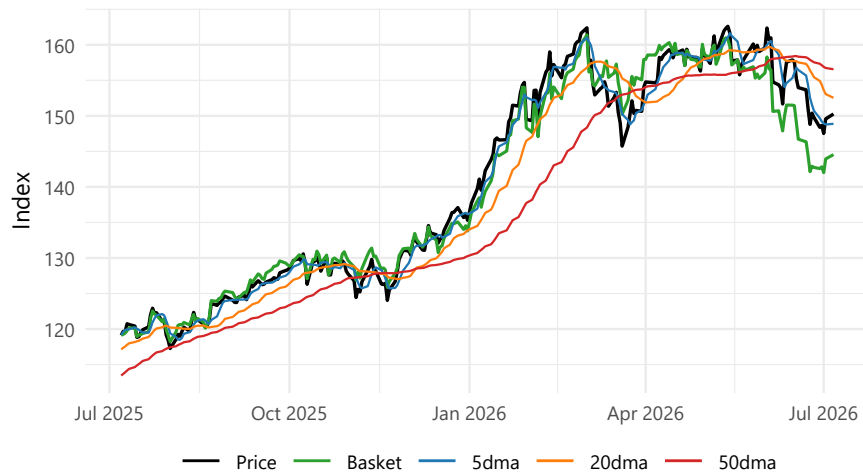
	ticker	PC_wt	name	category	etf_signal
21	IEMG	0.0425	iShares Core MSCI Emerging Markets ETF	Emerging Markets Equity	LONG
2	EEM	0.0424	iShares MSCI Emerging Index Fund	Emerging Markets Equity	FLAT
3	VWO	0.0422	Vanguard FTSE Emerging Markets ETF	Emerging Markets Equity	LONG
16	GEM	0.0422	Goldman Sachs ActiveBeta Emerging Marke...	Emerging Markets Equity	FLAT
20	ESGE	0.0422	iShares ESG Aware MSCI EM ETF	Emerging Markets Equity	FLAT
23	SPEM	0.0422	State Street SPDR Portfolio Emerging Ma...	Emerging Markets Equity	FLAT
22	SCHE	0.0421	Schwab Emerging Markets Equity ETF	Emerging Markets Equity	LONG
18	XSOE	0.0420	WisdomTree Emerging Markets Ex-State Ow...	Emerging Markets Equity	FLAT
5	AAXJ	0.0418	iShares MSCI All Country Asia ex Japan ...	Asia ex-Japan Equity	FLAT
15	EMGF	0.0418	iShares Emerging Markets Equity Factor ...	Emerging Markets Equity	FLAT
11	EEMA	0.0414	iShares MSCI Emerging Markets Asia ETF	Emerging Markets Equity	FLAT
14	GMF	0.0412	State Street SPDR S&P Emerging Asia Pac...	Asia Equity	FLAT
12	PXH	0.0409	Invesco RAFI Emerging Markets ETF	Emerging Markets Equity	FLAT
17	FNDE	0.0409	Schwab Fundamental Emerging Markets Equ...	Emerging Markets Equity	FLAT
19	EEMV	0.0405	iShares MSCI Emerging Markets Min Vol F...	Emerging Markets Equity	FLAT
8	DGS	0.0402	WisdomTree Emerging Market SmallCap Fund	Emerging Markets Equity	FLAT
10	AIA	0.0402	iShares Asia 50 ETF	Asia Equity	FLAT
7	DEM	0.0396	WisdomTree Emerging Markets High Divide...	Emerging Markets Equity	FLAT
6	EWX	0.0392	State Street SPDR S&P Emerging Markets ...	Emerging Markets Equity	LONG
4	FEM	0.0388	First Trust Emerging Markets AlphaDEX F...	Emerging Markets Equity	FLAT
13	DVYE	0.0365	iShares Emerging Markets Dividend Index...	Emerging Markets Equity	FLAT
9	EWI	0.0361	iShares MSCI Taiwan ETF	Taiwan Equity	LONG
25	FEMS	0.0349	First Trust Emerging Markets Small Cap ...	Emerging Markets Equity	FLAT
1	EWY	0.0341	iShares MSCI South Korea ETF	South Korea Equity	FLAT
24	SDIV	0.0338	Global X SuperDividend ETF	Global Dividend Equity	FLAT

Cluster 10 – Materials

Members: 21 **Var explained:** 0.702 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-07 – filter **PASS** (252d at date = +44.4%)
- Latest sell_cross: 2026-06-09
- Current 252d return: +24.8% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-06-09 more recent than last effective buy 2026-04-07)



Category breakdown

category	n_members	abs_weight
Materials	13	0.6545
MLP / Energy Infrastructure	4	0.1687
Energy	1	0.0502
Consumer Staples	1	0.0493
Norway Equity	1	0.0438
Industrial Metals	1	0.0336

Replication basket – AMLP+URA+GUNR (cor=0.96) basket lead at lag 1: 0.043

ticker	rep_wt	name	category	etf_signal
AMLP	0.109	Alerian MLP ETF	MLP / Energy Infrastr...	FLAT
URA	0.082	Global X Uranium ETF	Materials	FLAT
GUNR	0.776	FlexShares Global Upstream Natural Re...	Materials	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

ticker	PC_wt	name	category	etf_signal
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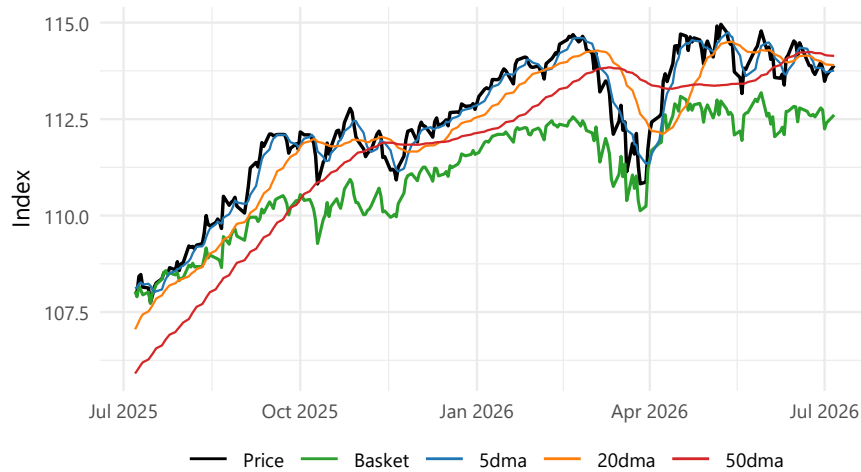
15	GNR	0.0550	State Street SPDR S&P Global Natural Re...	Materials	FLAT
14	GUNR	0.0543	FlexShares Global Upstream Natural Reso...	Materials	FLAT
10	MXI	0.0529	iShares Global Materials ETF	Materials	FLAT
11	IYM	0.0529	iShares U.S. Basic Materials ETF	Materials	FLAT
21	NANR	0.0525	State Street SPDR S&P North American Na...	Materials	FLAT
13	VAW	0.0514	Vanguard Materials ETF	Materials	LONG
16	PICK	0.0509	iShares MSCI Global Select Metals & Min...	Materials	FLAT
12	RSPM	0.0506	Invesco S&P 500 Equal Weight Materials ...	Materials	LONG
1	XLB	0.0505	State Street Materials Select Sector SP...	Materials	LONG
20	IGE	0.0502	iShares North American Natural Resource...	Energy	FLAT
4	XME	0.0499	State Street SPDR S&P Metals & Mining E...	Materials	FLAT
17	SLX	0.0498	VanEck Steel ETF	Materials	FLAT
6	MOO	0.0493	VanEck Agribusiness ETF	Consumer Staples	FLAT
3	COPX	0.0472	Global X Copper Miners ETF	Materials	FLAT
18	NORW	0.0438	Global X MSCI Norway ETF	Norway Equity	FLAT
5	EMLP	0.0436	First Trust North American Energy Infra...	MLP / Energy Infrastruct...	LONG
9	MLPX	0.0436	Global X MLP & Energy Infrastructure ETF	MLP / Energy Infrastruct...	LONG
7	AMLP	0.0413	Alerian MLP ETF	MLP / Energy Infrastruct...	FLAT
8	MLPA	0.0401	Global X MLP ETF	MLP / Energy Infrastruct...	LONG
2	URA	0.0367	Global X Uranium ETF	Materials	FLAT
19	DBB	0.0336	Invesco DB Base Metals Fund	Industrial Metals	FLAT

Cluster 11 – High Yield Bond

Members: 21 **Var explained:** 0.765 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-06-17 – filter **PASS** (252d at date = +7.3%)
- Latest sell_cross: 2026-06-26
- Current 252d return: +5.2% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-06-26 more recent than last effective buy 2026-06-17)



Category breakdown

category	n_members	abs_weight
High Yield Bond	10	0.5156
Preferred Stock	7	0.3059
Emerging Markets Fixed Income	4	0.1785

Replication basket – HYG (cor=0.95) basket lead at lag 1: 0.047

ticker	rep_wt	name	category	etf_signal
HYG	0.916	iShares iBoxx \$ High Yield Corporate ...	High Yield Bond	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
1	HYG	0.0526	iShares iBoxx \$ High Yield Corporate Bo...	High Yield Bond	LONG
13	JNK	0.0526	State Street SPDR Bloomberg High Yield ...	High Yield Bond	LONG
19	HYLB	0.0524	Xtrackers USD High Yield Corporate Bond...	High Yield Bond	LONG
18	BBHY	0.0523	JPMorgan BetaBuilders USD High Yield Co...	High Yield Bond	LONG
16	SHYG	0.0521	iShares 0-5 Year High Yield Corporate B...	High Yield Bond	LONG

14	SJNK	0.0520	State Street SPDR Bloomberg Short Term ...	High Yield Bond	LONG
15	ANGL	0.0516	VanEck Fallen Angel High Yield Bond ETF	High Yield Bond	LONG
17	FALN	0.0516	iShares Fallen Angels USD Bond ETF	High Yield Bond	LONG
6	HYS	0.0511	PIMCO 0-5 Year High Yield Corporat Bond...	High Yield Bond	LONG
9	PFF	0.0479	iShares Preferred and Income Securities...	Preferred Stock	FLAT
3	HYLS	0.0473	First Trust Tactical High Yield ETF	High Yield Bond	FLAT
2	EMB	0.0468	iShares J.P. Morgan USD Emerging Market...	Emerging Markets Fixed I...	FLAT
21	VWOB	0.0467	Vanguard Emerging Markets Government Bo...	Emerging Markets Fixed I...	LONG
11	PFXF	0.0461	VanEck Preferred Securities ex Financia...	Preferred Stock	FLAT
20	PCY	0.0458	Invesco Emerging Markets Sovereign Debt...	Emerging Markets Fixed I...	LONG
7	PGX	0.0445	Invesco Preferred ETF	Preferred Stock	FLAT
10	PSK	0.0441	State Street SPDR ICE Preferred Securit...	Preferred Stock	FLAT
4	FPE	0.0433	First Trust Preferred Securities and In...	Preferred Stock	FLAT
5	PGF	0.0431	Invesco Financial Preferred ETF	Preferred Stock	FLAT
12	HYEM	0.0393	VanEck Emerging Markets High Yield Bond...	Emerging Markets Fixed I...	FLAT
8	VRP	0.0368	Invesco Variable Rate Preferred ETF	Preferred Stock	LONG

Cluster 12 – Healthcare

Members: 17 **Var explained:** 0.760 **Signal:** LONG since 2026-05-06 (40 days, 13.3%)

Signal logic:

- Latest buy_cross: 2026-05-06 – filter **PASS** (252d at date = +13.8%)
- Latest sell_cross: 2026-04-29
- Current 252d return: +28.3% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-05-06 > last sell at 2026-04-29)



Category breakdown

category	n_members	abs_weight
Healthcare	16	0.9418
U.S. Small Cap Healthcare	1	0.0582

Replication basket – VHT (cor=0.93) basket lead at lag 1: 0.000

ticker	rep_wt	name	category	etf_signal
VHT	1.095	Vanguard Health Care ETF	Healthcare	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
15	VHT	0.0654	Vanguard Health Care ETF	Healthcare	LONG
16	FHLC	0.0653	Fidelity MSCI Health Care Index ETF	Healthcare	LONG
9	FXH	0.0646	First Trust Health Care AlphaDEX	Healthcare	LONG
14	RSPH	0.0640	Invesco S&P 500 Equal Weight Health Car...	Healthcare	LONG
13	IYH	0.0638	iShares U.S. Healthcare ETF	Healthcare	LONG
10	IXJ	0.0623	iShares Global Healthcare ETF	Healthcare	FLAT

1	XLV	0.0621	State Street Health Care Select Sector ...	Healthcare	LONG
3	IBB	0.0604	iShares Biotechnology ETF	Healthcare	LONG
6	FBT	0.0596	First Trust Amex Biotech Index Fund	Healthcare	FLAT
8	PSCH	0.0582	Invesco S&P SmallCap Health Care ETF	U.S. Small Cap Healthcare	LONG
7	XHE	0.0574	State Street SPDR S&P Health Care Equip...	Healthcare	LONG
12	IHI	0.0571	iShares U.S. Medical Devices ETF	Healthcare	FLAT
17	PPH	0.0543	VanEck Pharmaceutical ETF	Healthcare	LONG
5	PTH	0.0542	Invesco Dorsey Wright Healthcare Moment...	Healthcare	LONG
4	XBI	0.0534	State Street SPDR S&P Biotech ETF	Healthcare	LONG
2	ARKG	0.0508	ARK Genomic Revolution ETF	Healthcare	FLAT
11	IHF	0.0472	iShares U.S. Health Care Providers ETF	Healthcare	LONG

Cluster 13 – China Equity

Members: 15 **Var explained:** 0.799 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-15 – filter **PASS** (252d at date = +24.3%)
- Latest sell_cross: 2026-05-20
- Current 252d return: +5.6% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-20 more recent than last effective buy 2026-04-15)



Category breakdown

category	n_members	abs_weight
China Equity	13	0.8765
Emerging Markets Equity	1	0.0643
Hong Kong Equity	1	0.0592

Replication basket – KWEB (cor=0.92) basket lead at lag 1: 0.011

ticker	rep_wt	name	category	etf_signal
KWEB	0.563	KraneShares CSI China Internet ETF	China Equity	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

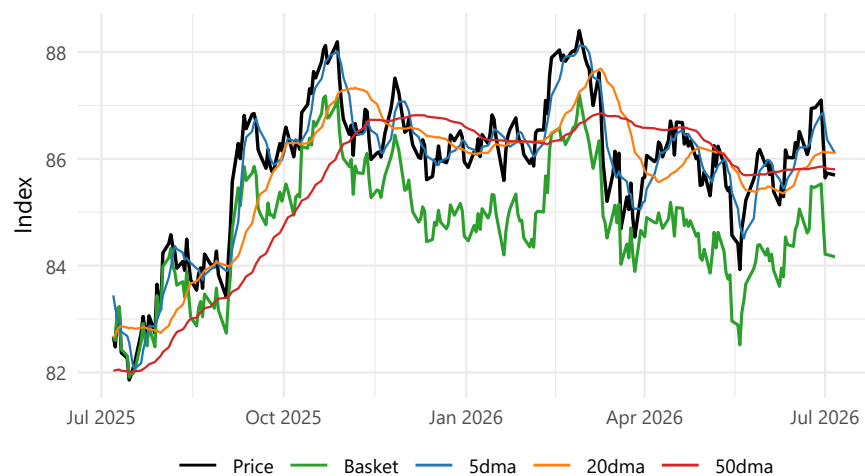
	ticker	PC_wt	name	category	etf_signal
11	CXSE	0.0738	WisdomTree China ex-State-Owned Enterpr...	China Equity	FLAT
9	GXC	0.0736	State Street SPDR S&P China ETF	China Equity	FLAT
1	MCHI	0.0730	iShares MSCI China ETF	China Equity	FLAT
2	FXI	0.0712	iShares China Large-Cap ETF	China Equity	FLAT
5	CQQQ	0.0712	Invesco China Technology ETF	China Equity	FLAT
7	CHIQ	0.0705	Global X MSCI China Consumer Discretion...	China Equity	FLAT
4	KWEB	0.0680	KraneShares CSI China Internet ETF	China Equity	FLAT
6	ASHR	0.0668	Xtrackers Harvest CSI 300 China A-Share...	China Equity	LONG
8	CNYA	0.0661	iShares MSCI China A ETF	China Equity	LONG
13	KGRN	0.0652	KraneShares MSCI China Clean Technology...	China Equity	FLAT
3	EMQQ	0.0643	EMQQ The Emerging Markets Internet ETF	Emerging Markets Equity	FLAT
10	KBA	0.0596	KraneShares Boseri MSCI China A 50 Conn...	China Equity	FLAT
12	EWK	0.0592	iShares MSCI Hong Kong Index Fund	Hong Kong Equity	FLAT
14	CNXT	0.0592	VanEck ChiNext Innovators ETF	China Equity	FLAT
15	KURE	0.0583	KraneShares MSCI All China Health Care ...	China Equity	LONG

Cluster 14 – Fixed Income

Members: 15 **Var explained:** 0.928 **Signal:** LONG since 2026-06-15 (13 days, -0.4%)

Signal logic:

- Latest buy_cross: 2026-06-15 – filter **PASS** (252d at date = +3.9%)
- Latest sell_cross: 2026-05-13
- Current 252d return: +2.5% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-06-15 > last sell at 2026-05-13)



Category breakdown

category	n_members	abs_weight
Investment Grade Corporate Bond	7	0.4707
Fixed Income	6	0.4009
Active Fixed Income	1	0.0665
Taxable Municipal Bond	1	0.0619

Replication basket – TLT (cor=0.97) basket lead at lag 1: -0.041

ticker	rep_wt	name	category	etf_signal
TLT	0.694	iShares 20+ Year Treasury Bond ETF	Fixed Income	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
9	BLV	0.0687	Vanguard Long-Term Bond ETF	Fixed Income	LONG
7	SPLB	0.0678	State Street SPDR Portfolio Long Term C...	Investment Grade Corpora...	LONG
8	IGLB	0.0678	iShares 10 Year Investment Grade Corpor...	Investment Grade Corpora...	LONG
10	VCLT	0.0676	Vanguard Long-Term Corporate Bond ETF	Investment Grade Corpora...	LONG
2	LQD	0.0675	iShares iBoxx \$ Investment Grade Corpor...	Investment Grade Corpora...	LONG
11	USIG	0.0672	iShares Broad USD Investment Grade Corp...	Investment Grade Corpora...	LONG
6	QLTA	0.0671	iShares Aaa A Rated Corporate Bond ETF	Investment Grade Corpora...	LONG
12	TLH	0.0668	iShares 10-20 Year Treasury Bond ETF	Fixed Income	LONG
14	SPTL	0.0667	State Street SPDR Portfolio Long Term T...	Fixed Income	LONG
15	VGLT	0.0666	Vanguard Long-Term Treasury ETF	Fixed Income	LONG
1	TLT	0.0665	iShares 20+ Year Treasury Bond ETF	Fixed Income	LONG
3	FLCO	0.0665	Franklin Investment Grade Corporate ETF	Active Fixed Income	LONG
5	CORP	0.0657	PIMCO Investment Grade Corporate Bond I...	Investment Grade Corpora...	LONG
13	EDV	0.0656	Vanguard Extended Duration Treasury ETF	Fixed Income	LONG
4	BAB	0.0619	Invesco Taxable Municipal Bond ETF	Taxable Municipal Bond	LONG

Cluster 15 – Real Estate

Members: 14 **Var explained:** 0.961 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-10 – filter **PASS** (252d at date = +21.2%)
- Latest sell_cross: 2026-06-03
- Current 252d return: +13.1% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-06-03 more recent than last effective buy 2026-04-10)



Category breakdown

category	n_members	abs_weight
Real Estate	14	1

Replication basket – VNQ (cor=0.99) basket lead at lag 1: -0.000

ticker	rep_wt	name	category	etf_signal
VNQ	0.97	Vanguard Real Estate ETF	Real Estate	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
8	VNQ	0.0725	Vanguard Real Estate ETF	Real Estate	FLAT
9	FREL	0.0725	Fidelity MSCI Real Estate Index ETF	Real Estate	LONG
11	SCHH	0.0725	Schwab U.S. REIT ETF	Real Estate	FLAT
2	IYR	0.0724	iShares U.S. Real Estate ETF	Real Estate	LONG
10	USRT	0.0724	iShares Core U.S. REIT ETF	Real Estate	FLAT
6	RWR	0.0723	State Street SPDR Dow Jones REIT ETF	Real Estate	FLAT
13	PSR	0.0722	Invesco Active U.S. Real Estate Fund	Real Estate	FLAT

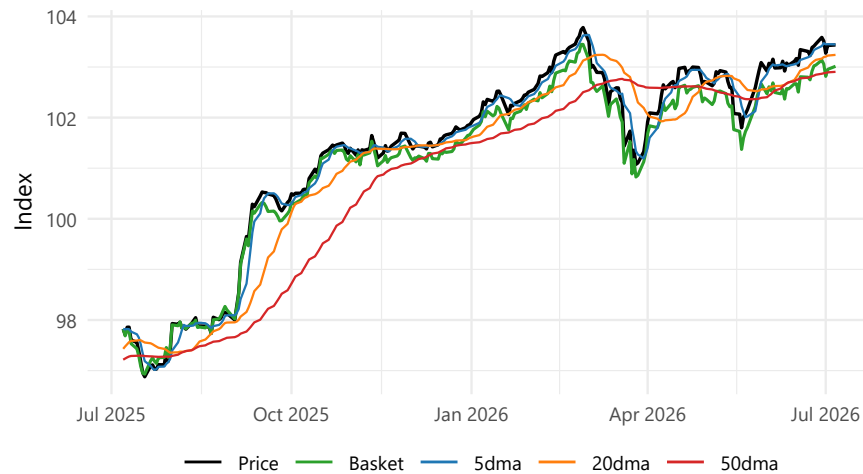
1	XLRE	0.0719	State Street Real Estate Select Sector ...	Real Estate	LONG
5	ICF	0.0719	iShares Select U.S. REIT ETF	Real Estate	FLAT
7	REET	0.0716	iShares Global REIT ETF	Real Estate	FLAT
3	RWO	0.0711	State Street SPDR Dow Jones Global Real...	Real Estate	LONG
4	REZ	0.0697	iShares Residential and Multisector Rea...	Real Estate	FLAT
14	NURE	0.0688	Nuveen Short-Term REIT ETF	Real Estate	LONG
12	INDS	0.0682	Pacer Industrial Real Estate ETF	Real Estate	LONG

Cluster 16 – Municipal Bond

Members: 13 **Var explained:** 0.819 **Signal:** LONG since 2026-05-28 (25 days, 0.5%)

Signal logic:

- Latest buy_cross: 2026-05-28 – filter **PASS** (252d at date = +5.9%)
- Latest sell_cross: 2026-05-15
- Current 252d return: +5.7% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-05-28 > last sell at 2026-05-15)



Category breakdown

category	n_members	abs_weight
Municipal Bond	13	1

Replication basket – MUB (cor=0.95) basket lead at lag 1: 0.177

ticker	rep_wt	name	category	etf_signal
MUB	0.904	iShares National Muni Bond ETF	Municipal Bond	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
13	VTEB	0.0817	Vanguard Tax-Exempt Bond ETF	Municipal Bond	LONG
11	MUB	0.0813	iShares National Muni Bond ETF	Municipal Bond	LONG
6	CMF	0.0802	iShares California Muni Bond ETF	Municipal Bond	LONG
9	TFI	0.0802	State Street SPDR Nuveen ICE Municipal ...	Municipal Bond	LONG
8	ITM	0.0800	VanEck Intermediate Muni ETF	Municipal Bond	LONG
1	FMB	0.0794	First Trust Managed Municipal ETF	Municipal Bond	LONG
7	NYF	0.0793	iShares New York Muni Bond ETF	Municipal Bond	LONG
5	PZA	0.0780	Invesco National AMT-Free Municipal Bon...	Municipal Bond	LONG
4	MUNI	0.0769	PIMCO Intermediate Municipal Bond Activ...	Municipal Bond	LONG
2	HYD	0.0747	VanEck High Yield Muni ETF	Municipal Bond	LONG
3	HYMB	0.0721	State Street SPDR Nuveen ICE High Yield...	Municipal Bond	LONG
12	SUB	0.0695	iShares Short-Term National Muni Bond E...	Municipal Bond	FLAT
10	SHM	0.0667	State Street SPDR Nuveen ICE Short Term...	Municipal Bond	LONG

Cluster 17 – Gold

Members: 12 **Var explained:** 0.861 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-05-12 – filter **PASS** (252d at date = +75.1%)
- Latest sell_cross: 2026-05-20
- Current 252d return: +31.9% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-20 more recent than last effective buy 2026-05-12)



Category breakdown

category	n_members	abs_weight
Materials	5	0.4157
Gold	3	0.2502
Precious Metals	2	0.1724
Silver	2	0.1617

Replication basket – GLD+IAU+SLV (cor=0.95) basket lead at lag 1: -0.009

ticker	rep_wt	name	category	etf_signal
GLD	0.009	SPDR Gold Shares	Gold	FLAT
IAU	0.764	iShares Gold Trust Shares	Gold	FLAT
SLV	0.411	iShares Silver Trust	Silver	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
11	DBP	0.0866	Invesco DB Precious Metals Fund	Precious Metals	FLAT
7	GLTR	0.0859	abrdn Physical Precious Metals Basket S...	Precious Metals	FLAT

2	GDXJ	0.0841	VanEck Junior Gold Miners ETF	Materials	FLAT
1	GDX	0.0840	VanEck Gold Miners ETF	Materials	FLAT
3	GLD	0.0835	SPDR Gold Shares	Gold	FLAT
9	IAU	0.0834	iShares Gold Trust Shares	Gold	FLAT
10	SGOL	0.0833	abrdn Physical Gold Shares ETF	Gold	FLAT
12	GOAU	0.0831	US Global GO Gold and Precious Metal Mi...	Materials	FLAT
6	SIL	0.0828	Global X Silver Miners ETF	Materials	FLAT
5	SILJ	0.0818	Amplify Junior Silver Miners ETF	Materials	FLAT
4	SLV	0.0809	iShares Silver Trust	Silver	LONG
8	SIVR	0.0808	abrdn Physical Silver Shares ETF	Silver	LONG

Cluster 18 – Consumer Staples

Members: 11 **Var explained:** 0.745 **Signal:** LONG since 2026-06-26 (5 days, -0.9%)

Signal logic:

- Latest buy_cross: 2026-06-26 – filter **PASS** (252d at date = +9.9%)
- Latest sell_cross: 2026-05-29
- Current 252d return: +6.9% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-06-26 > last sell at 2026-05-29)



Category breakdown

category	n_members	abs_weight
Consumer Staples	6	0.5733
Utilities	4	0.3520
U.S. Small Cap Consumer Staples	1	0.0747

Replication basket – XLU+XLP (cor=0.99) basket lead at lag 1: -0.013

ticker	rep_wt	name	category	etf_signal
XLU	0.364	State Street Utilities Select Sector ...	Utilities	LONG
XLP	0.588	State Street Consumer Staples Select ...	Consumer Staples	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
5	VDC	0.0983	Vanguard Consumer Staples ETF	Consumer Staples	LONG
6	FSTA	0.0983	Fidelity MSCI Consumer Staples Index ETF	Consumer Staples	LONG
1	XLP	0.0975	State Street Consumer Staples Select Se...	Consumer Staples	LONG
4	IYK	0.0949	iShares U.S. Consumer Staples ETF	Consumer Staples	LONG
3	KXI	0.0946	iShares Global Consumer Staples ETF	Consumer Staples	LONG
11	PBJ	0.0897	Invesco Food & Beverage ETF	Consumer Staples	FLAT
7	IDU	0.0890	iShares U.S. Utilities ETF	Utilities	LONG
9	FUTY	0.0878	Fidelity MSCI Utilities Index ETF	Utilities	LONG
8	VPU	0.0877	Vanguard Utilities ETF	Utilities	LONG
2	XLU	0.0874	State Street Utilities Select Sector SP...	Utilities	LONG
10	PSCC	0.0747	Invesco S&P SmallCap Consumer Staples E...	U.S. Small Cap Consumer ...	LONG

Cluster 19 – Broad Commodities

Members: 11 **Var explained:** 0.771 **Signal:** LONG since 2026-06-03 (21 days, -13.3%)

Signal logic:

- Latest buy_cross: 2026-06-03 – filter **PASS** (252d at date = +48.1%)
- Latest sell_cross: 2026-05-27
- Current 252d return: +23.8% – filter **ALLOWS** new entries today

- Decision: **LONG** (effective buy at 2026-06-03 > last sell at 2026-05-27)



Category breakdown

category	n_members	abs_weight
Broad Commodities	6	0.6173
Oil	2	0.1979
Energy Commodities	1	0.1019
Agricultural Commodities	1	0.0500
Natural Gas	1	0.0329

Replication basket – PDBC (cor=0.98) basket lead at lag 1: 0.075

ticker	rep_wt	name	category	etf_signal
PDBC	1.121	Invesco Optimum Yield Diversified Com...	Broad Commodities	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
6	GSG	0.1056	iShares GSCI Commodity-Indexed Trust Fu...	Broad Commodities	FLAT

2	DBC	0.1055	Invesco DB Commodity Index Tracking Fund	Broad Commodities	LONG
8	PDBC	0.1055	Invesco Optimum Yield Diversified Commo...	Broad Commodities	LONG
9	COMT	0.1052	iShares GSCI Commodity Dynamic Roll Str...	Broad Commodities	FLAT
10	DBE	0.1019	Invesco DB Energy Fund	Energy Commodities	LONG
1	USO	0.0995	United States Oil Fund	Oil	LONG
5	DBO	0.0984	Invesco DB Oil Fund	Oil	FLAT
7	DJP	0.0982	iPath Bloomberg Commodity Index Total R...	Broad Commodities	LONG
4	FTGC	0.0972	First Trust Global Tactical Commodity S...	Broad Commodities	FLAT
3	DBA	0.0500	Invesco DB Agriculture Fund	Agricultural Commodities	FLAT
11	UNG	0.0329	United States Natural Gas Fund LP	Natural Gas	FLAT

Cluster 20 – India Equity

Members: 10 **Var explained:** 0.405 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-05-06 – filter **PASS** (252d at date = +4.7%)
- Latest sell_cross: 2026-05-13
- Current 252d return: -13.5% – filter **BLOCKS** new entries today
- Decision: **FLAT** (last sell 2026-05-13 more recent than last effective buy 2026-05-06)



Category breakdown

category	n_members	abs_weight
India Equity	3	0.4678
Malaysia Equity	1	0.1127
Indonesia Equity	1	0.1042
Philippines Equity	1	0.1011
Vietnam Equity	1	0.0749
Turkey Equity	1	0.0611
U.S. Dividend Equity	1	0.0548
Platinum	1	0.0235

Replication basket – NOBL+INDA+PPLT (cor=0.91) basket lead at lag 1: -0.006

ticker	rep_wt	name	category	etf_signal
NOBL	0.080	ProShares S&P 500 Dividend Aristocrat...	U.S. Dividend Equity	FLAT
INDA	0.660	iShares MSCI India ETF	India Equity	FLAT
PPLT	0.029	abrdn Physical Platinum Shares ETF	Platinum	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
1	INDA	0.1573	iShares MSCI India ETF	India Equity	FLAT
2	INDY	0.1556	iShares India 50 ETF	India Equity	FLAT
3	EPI	0.1548	WisdomTree India Earnings Fund	India Equity	FLAT
8	EWM	0.1127	iShares MSCI Malaysia Index Fund	Malaysia Equity	FLAT
6	EIDO	0.1042	iShares MSCI Indonesia ETF	Indonesia Equity	FLAT
7	EPHE	0.1011	iShares MSCI Philippines ETF	Philippines Equity	FLAT
10	VNM	0.0749	VanEck Vietnam ETF	Vietnam Equity	FLAT
9	TUR	0.0611	iShares MSCI Turkey ETF	Turkey Equity	FLAT
4	NOBL	0.0548	ProShares S&P 500 Dividend Aristocrats ...	U.S. Dividend Equity	FLAT
5	PPLT	0.0235	abrdn Physical Platinum Shares ETF	Platinum	FLAT

Cluster 21 – Inflation-Protected Fixed Income

Members: 10 **Var explained:** 0.905 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-05-27 – filter **PASS** (252d at date = +4.8%)
- Latest sell_cross: 2026-07-02
- Current 252d return: +1.3% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-07-02 more recent than last effective buy 2026-05-27)



Category breakdown

category	n_members	abs_weight
Inflation-Protected Fixed Income	10	1

Replication basket – VTIP+SCHP (cor=0.99) basket lead at lag 1: -0.013

ticker	rep_wt	name	category	etf_signal
VTIP	0.269	Vanguard Short-Term Inflation-Protect...	Inflation-Protected F...	FLAT
SCHP	0.759	Schwab U.S. TIPS ETF	Inflation-Protected F...	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
4	TDTF	0.1035	FlexShares iBoxx 5 Year Target Duration...	Inflation-Protected Fixe...	FLAT

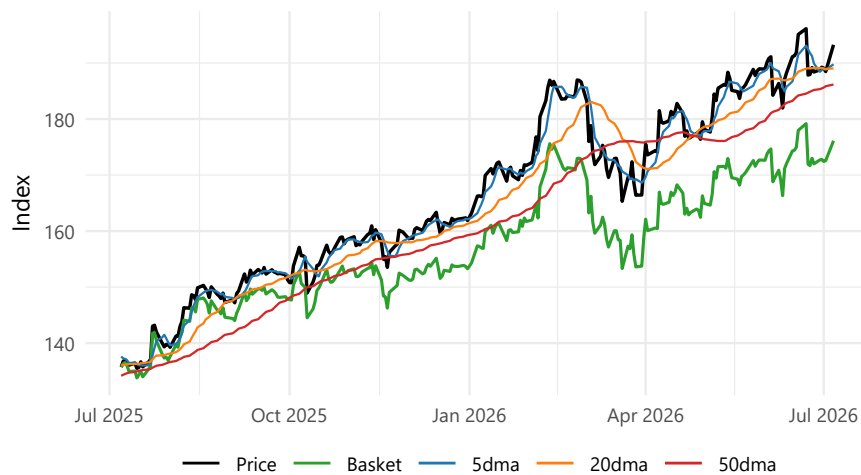
6	TIPX	0.1034	State Street SPDR Bloomberg 1a??10 Year...	Inflation-Protected Fixe...	FLAT
3	TIP	0.1030	iShares TIPS Bond ETF	Inflation-Protected Fixe...	FLAT
8	SCHP	0.1023	Schwab U.S. TIPS ETF	Inflation-Protected Fixe...	FLAT
5	TDDT	0.1008	FlexShares iBoxx 3 Year Target Duration...	Inflation-Protected Fixe...	FLAT
7	SPIP	0.1000	State Street SPDR Portfolio TIPS ETF	Inflation-Protected Fixe...	FLAT
9	STIP	0.0998	iShares 0-5 Year TIPS Bond ETF	Inflation-Protected Fixe...	FLAT
2	STPZ	0.0996	PIMCO 1-5 Year U.S. TIPS Index Exchange...	Inflation-Protected Fixe...	FLAT
10	VTIP	0.0990	Vanguard Short-Term Inflation-Protected...	Inflation-Protected Fixe...	FLAT
1	LTPZ	0.0885	Pimco 15 Year U.S. TIPS Index Exchange...	Inflation-Protected Fixe...	FLAT

Cluster 22 – Japan Equity

Members: 9 **Var explained:** 0.887 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-09 – filter **PASS** (252d at date = +57.2%)
- Latest sell_cross: 2026-06-11
- Current 252d return: +40.3% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-06-11 more recent than last effective buy 2026-04-09)



Category breakdown

category	n_members	abs_weight
Japan Equity	7	0.7803
Asia-Pacific Equity	2	0.2197

Replication basket – EWJ (cor=0.97) basket lead at lag 1: -0.041

ticker	rep_wt	name	category	etf_signal
EWJ	0.91	iShares MSCI Japan Index Fund	Japan Equity	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

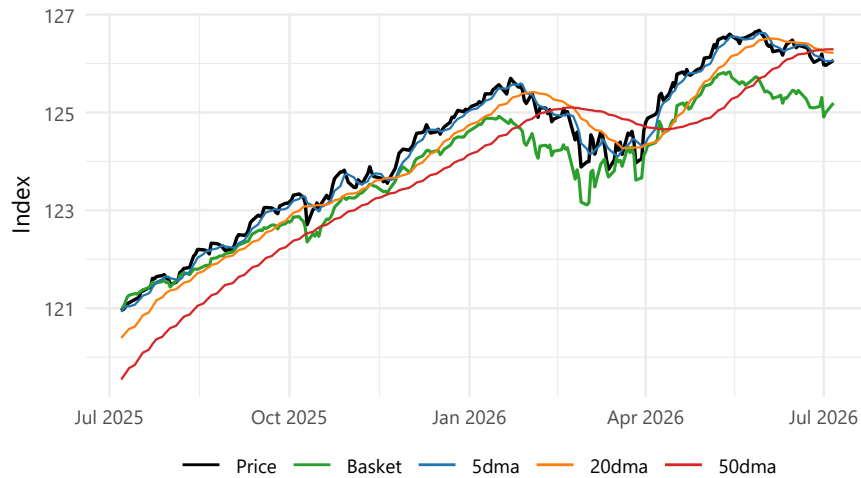
	ticker	PC_wt	name	category	etf_signal
1	EWJ	0.1142	iShares MSCI Japan Index Fund	Japan Equity	FLAT
8	BBJP	0.1142	JPMorgan BetaBuilders Japan ETF	Japan Equity	FLAT
9	HEWJ	0.1118	iShares Currency Hedged MSCI Japan ETF	Japan Equity	LONG
6	DBJP	0.1117	Xtrackers MSCI Japan Hedged Equity ETF	Japan Equity	LONG
3	IPAC	0.1114	iShares Core MSCI Pacific ETF	Asia-Pacific Equity	LONG
5	FLJH	0.1109	Franklin FTSE Japan Hedged ETF	Japan Equity	FLAT
2	DXJ	0.1097	WisdomTree Japan Hedged Equity Fund	Japan Equity	FLAT
4	VPL	0.1083	Vanguard FTSE Pacific ETF	Asia-Pacific Equity	FLAT
7	EWJV	0.1079	iShares MSCI Japan Value ETF	Japan Equity	LONG

Cluster 23 – Bank Loan / Floating Rate

Members: 9 **Var explained:** 0.557 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-07 – filter **PASS** (252d at date = +6.5%)
- Latest sell_cross: 2026-06-23
- Current 252d return: +4.2% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-06-23 more recent than last effective buy 2026-04-07)



Category breakdown

category	n_members	abs_weight
Bank Loan / Floating Rate	3	0.3624
Interest Rate Hedged Fixed Income	3	0.3243
Floating Rate Fixed Income	3	0.3133

Replication basket – FLOT+BKLN+SRLN+FLRN+FLTR (cor=0.89) basket lead at lag 1: -0.047

ticker	rep_wt	name	category	etf_signal
FLOT	0.292	iShares Floating Rate Bond ETF	Floating Rate Fixed I...	FLAT
BKLN	0.214	Invesco Senior Loan ETF	Bank Loan / Floating ...	FLAT
SRLN	0.303	State Street Blackstone Senior Loan E...	Bank Loan / Floating ...	FLAT
FLRN	0.127	State Street SPDR Bloomberg Investmen...	Floating Rate Fixed I...	FLAT
FLTR	0.035	VanEck IG Floating Rate ETF	Floating Rate Fixed I...	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
2	SRLN	0.1250	State Street Blackstone Senior Loan ETF	Bank Loan / Floating Rate	FLAT

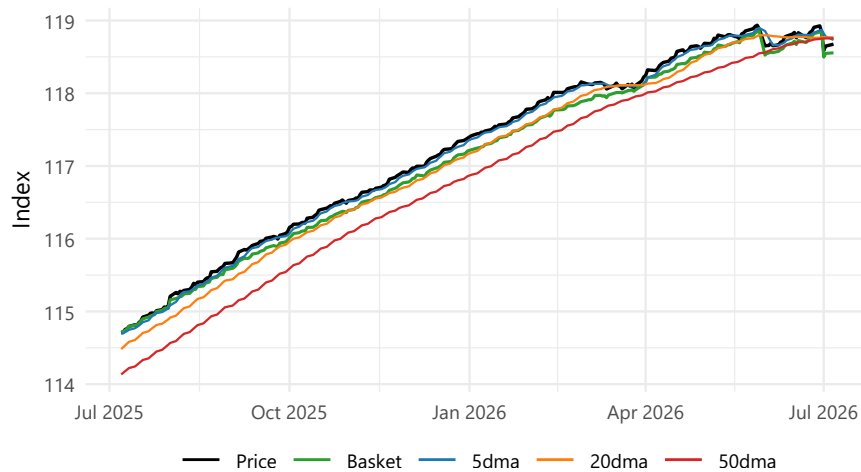
3	BKLN	0.1209	Invesco Senior Loan ETF	Bank Loan / Floating Rate	FLAT
1	FTSL	0.1165	First Trust Senior Loan Fund	Bank Loan / Floating Rate	FLAT
5	LQDH	0.1143	iShares Interest Rate Hedged Corporate ...	Interest Rate Hedged Fix...	FLAT
7	FLOT	0.1084	iShares Floating Rate Bond ETF	Floating Rate Fixed Inco...	FLAT
6	IGBH	0.1056	iShares Interest Rate Hedged Long-Term ...	Interest Rate Hedged Fix...	FLAT
4	IGHG	0.1044	ProShares Investment Grade-Interest Rat...	Interest Rate Hedged Fix...	FLAT
8	FLRN	0.1039	State Street SPDR Bloomberg Investment ...	Floating Rate Fixed Inco...	FLAT
9	FLTR	0.1011	VanEck IG Floating Rate ETF	Floating Rate Fixed Inco...	FLAT

Cluster 24 – Cash / Ultra-Short Fixed Income

Members: 9 Var explained: 0.457 Signal: FLAT

Signal logic:

- Latest buy_cross: 2022-10-28 – filter **BLOCKED** (252d at date = -0.5%; needs > 0)
- Latest effective buy: 2022-11-29 – delayed re-arm from blocked cross (252d at fire = +0.0%)
- Latest sell_cross: 2026-06-03
- Current 252d return: +3.5% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-06-03 more recent than last effective buy 2022-11-29)



Category breakdown

category	n_members	abs_weight
Cash / Ultra-Short Fixed Income	9	1

Replication basket – SHV+MINT+GBIL+ICSH (cor=0.85) basket lead at lag 1: 0.066

ticker	rep_wt	name	category	etf_signal
SHV	0.274	iShares 0-1 Year Treasury Bond ETF	Cash / Ultra-Short Fi...	FLAT
MINT	0.172	PIMCO Enhanced Short Maturity Active ...	Cash / Ultra-Short Fi...	FLAT
GBIL	0.122	Goldman Sachs Access Treasury 0-1 Yea...	Cash / Ultra-Short Fi...	FLAT
ICSH	0.403	iShares Ultra Short Duration Bond Act...	Cash / Ultra-Short Fi...	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
7	SHV	0.1306	iShares 0-1 Year Treasury Bond ETF	Cash / Ultra-Short Fixed...	FLAT
6	ICSH	0.1296	iShares Ultra Short Duration Bond Activ...	Cash / Ultra-Short Fixed...	FLAT
5	GSY	0.1181	Invesco Ultra Short Duration ETF	Cash / Ultra-Short Fixed...	FLAT
2	MINT	0.1126	PIMCO Enhanced Short Maturity Active Ex...	Cash / Ultra-Short Fixed...	FLAT
3	FTSM	0.1123	First Trust Enhanced Short Maturity ETF	Cash / Ultra-Short Fixed...	FLAT
4	NEAR	0.1103	iShares Short Duration Bond Active ETF	Cash / Ultra-Short Fixed...	LONG
8	BIL	0.1016	State Street SPDR Bloomberg 1-3 Month T...	Cash / Ultra-Short Fixed...	FLAT
1	VNLA	0.0951	Janus Henderson Short Duration Income E...	Cash / Ultra-Short Fixed...	FLAT
9	GBIL	0.0898	Goldman Sachs Access Treasury 0-1 Year ...	Cash / Ultra-Short Fixed...	FLAT

Cluster 25 – International Fixed Income

Members: 6 **Var explained:** 0.742 **Signal:** LONG since 2026-05-29 (24 days, -2.2%)

Signal logic:

- Latest buy_cross: 2026-05-29 – filter **PASS** (252d at date = +0.6%)
- Latest sell_cross: 2026-05-18
- Current 252d return: -4.5% – filter **BLOCKS** new entries today
- Decision: **LONG** (effective buy at 2026-05-29 > last sell at 2026-05-18)



Category breakdown

category	n_members	abs_weight
Currency	4	0.641
International Fixed Income	2	0.359

Replication basket – BWX (cor=0.93) basket lead at lag 1: 0.026

ticker	rep_wt	name	category	etf_signal
BWX	0.751	State Street SPDR Bloomberg Internati...	International Fixed I...	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
2	IGOV	0.1801	iShares International Treasury Bond ETF	International Fixed Inco...	FLAT
1	BWX	0.1790	State Street SPDR Bloomberg Internation...	International Fixed Inco...	FLAT
4	FXE	0.1702	Invesco CurrencyShares Euro Currency Tr...	Currency	FLAT
5	FXF	0.1649	Invesco CurrencyShares Swiss Franc Trust	Currency	FLAT
3	FXB	0.1610	Invesco CurrencyShares British Pound St...	Currency	FLAT
6	FXY	0.1448	Invesco CurrencyShares Japanese Yen Tru...	Currency	FLAT

Cluster 26 – Leveraged / Inverse**Members:** 2 **Var explained:** 0.625 **Signal:** FLAT**Signal logic:**

- Latest buy_cross: 2026-02-05 – filter **BLOCKED** (252d at date = -32.0%; needs > 0)
- Latest effective buy: 2025-04-08 – delayed re-arm from blocked cross (252d at fire = +0.4%)
- Latest sell_cross: 2026-07-06
- Current 252d return: -35.9% – filter **BLOCKS** new entries today
- Decision: **FLAT** (last sell 2026-07-06 more recent than last effective buy 2025-04-08)

**Category breakdown**

category	n_members	abs_weight
Leveraged / Inverse	1	0.5
Currency	1	0.5

Replication basket – SQQQ+UUP (cor=1.00) basket lead at lag 1: -0.023

ticker	rep_wt	name	category	etf_signal
SQQQ	0.506	ProShares UltraPro Short QQQ	Leveraged / Inverse	FLAT

UUP	0.495	Invesco DB USD Index Bullish Fund ETF	Currency	LONG
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Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

ticker	PC_wt	name	category	etf_signal
SQQQ	0.5	ProShares UltraPro Short QQQ	Leveraged / Inverse	FLAT
UUP	0.5	Invesco DB USD Index Bullish Fund ETF	Currency	LONG

Long ETF Positions (full)

Full reference list of currently-long ETFs. Sorted by cluster's NSS desc, then ETF NSS within the cluster – leading themes appear first, same-theme ETFs are grouped. The actionable buy list is the Position Sheet up front; this table is for cases where you want to inspect all member-level positions.

ticker	cluster	theme	entry	days	return	p_sell	nss
PTH	cluster_12	Healthcare	2026-06-17	11	15.3%	0.00	1.00
XBI	cluster_12	Healthcare	2026-06-16	12	18.8%	0.00	1.00
PSCH	cluster_12	Healthcare	2026-04-13	57	26.4%	0.00	1.00
IBB	cluster_12	Healthcare	2026-06-17	11	12.6%	0.00	1.00
IHF	cluster_12	Healthcare	2026-05-08	38	13.8%	0.00	1.00
FXH	cluster_12	Healthcare	2026-04-10	58	12.5%	0.00	1.00
XHE	cluster_12	Healthcare	2026-06-09	17	7.0%	0.00	1.00
VHT	cluster_12	Healthcare	2026-05-19	31	11.6%	0.00	1.00
FHLC	cluster_12	Healthcare	2026-05-19	31	11.6%	0.00	1.00
RSPH	cluster_12	Healthcare	2026-05-21	29	11.4%	0.00	1.00
IYH	cluster_12	Healthcare	2026-05-20	30	10.8%	0.00	1.00
XLV	cluster_12	Healthcare	2026-05-21	29	9.3%	0.00	1.00
PPH	cluster_12	Healthcare	2026-05-21	29	5.9%	0.00	1.00
KIE	cluster_7	U.S. Small Cap Equity	2026-06-10	16	9.9%	0.00	1.00
FXO	cluster_7	U.S. Small Cap Equity	2026-06-05	19	8.0%	0.00	1.00
SDVY	cluster_7	U.S. Small Cap Equity	2026-06-08	18	4.8%	0.00	1.00
FNCL	cluster_7	U.S. Small Cap Equity	2026-04-13	57	8.7%	0.00	1.00
IAT	cluster_7	U.S. Small Cap Equity	2026-05-20	30	11.5%	0.00	1.00
XLF	cluster_7	U.S. Small Cap Equity	2026-04-13	57	8.7%	0.00	1.00
ISMD	cluster_7	U.S. Small Cap Equity	2026-04-09	59	17.3%	0.00	1.00
IJS	cluster_7	U.S. Small Cap Equity	2026-04-10	58	10.2%	0.01	0.99
IAI	cluster_7	U.S. Small Cap Equity	2026-07-06	0	0.0%	0.79	-0.34
FXR	cluster_1	U.S. Mid Cap Equity	2026-06-12	14	3.3%	0.00	1.00
ITA	cluster_1	U.S. Mid Cap Equity	2026-05-26	27	9.3%	0.00	1.00
ITB	cluster_1	U.S. Mid Cap Equity	2026-06-09	17	5.2%	0.00	1.00
PSCD	cluster_1	U.S. Mid Cap Equity	2026-05-27	26	7.8%	0.00	1.00
PEJ	cluster_1	U.S. Mid Cap Equity	2026-05-26	27	7.5%	0.00	1.00
XHB	cluster_1	U.S. Mid Cap Equity	2026-05-29	24	8.4%	0.00	1.00
IWR	cluster_1	U.S. Mid Cap Equity	2026-04-10	58	9.9%	0.00	1.00
IMCB	cluster_1	U.S. Mid Cap Equity	2026-04-10	58	12.3%	0.01	0.99
SCHA	cluster_1	U.S. Mid Cap Equity	2026-04-09	59	13.7%	0.02	0.98
SIZE	cluster_1	U.S. Mid Cap Equity	2026-04-13	57	8.7%	0.02	0.98
IWP	cluster_1	U.S. Mid Cap Equity	2026-04-15	55	6.6%	0.02	0.98
PPA	cluster_1	U.S. Mid Cap Equity	2026-05-27	26	4.6%	0.02	0.97
IWO	cluster_1	U.S. Mid Cap Equity	2026-04-13	57	15.1%	0.03	0.96
IMCG	cluster_1	U.S. Mid Cap Equity	2026-04-09	59	17.7%	0.04	0.96
VTWG	cluster_1	U.S. Mid Cap Equity	2026-04-13	57	15.1%	0.04	0.96

VFMO	cluster_1	U.S. Mid Cap Equity	2026-04-08	60	13.4%	0.06	0.94
SCHM	cluster_1	U.S. Mid Cap Equity	2026-04-09	59	11.1%	0.06	0.94
XRT	cluster_1	U.S. Mid Cap Equity	2026-05-28	25	2.3%	0.07	0.93
IVOG	cluster_1	U.S. Mid Cap Equity	2026-04-09	59	9.8%	0.08	0.91
IJK	cluster_1	U.S. Mid Cap Equity	2026-04-09	59	9.8%	0.09	0.91
MDYG	cluster_1	U.S. Mid Cap Equity	2026-04-09	59	9.7%	0.11	0.89
PKB	cluster_1	U.S. Mid Cap Equity	2026-06-16	12	-0.4%	0.11	0.89
FV	cluster_1	U.S. Mid Cap Equity	2026-04-14	56	14.7%	0.12	0.88
VO	cluster_1	U.S. Mid Cap Equity	2026-04-13	57	7.9%	0.13	0.87
VCR	cluster_1	U.S. Mid Cap Equity	2026-07-01	2	-0.1%	0.02	0.78
FDIS	cluster_1	U.S. Mid Cap Equity	2026-07-01	2	0.0%	0.02	0.77
RSPD	cluster_1	U.S. Mid Cap Equity	2026-06-11	15	0.3%	0.26	0.74
FTXR	cluster_1	U.S. Mid Cap Equity	2026-04-17	53	9.7%	0.26	0.74
VOT	cluster_1	U.S. Mid Cap Equity	2026-04-14	56	10.1%	0.26	0.72
FXD	cluster_1	U.S. Mid Cap Equity	2026-05-27	26	2.0%	0.34	0.66
RPG	cluster_1	U.S. Mid Cap Equity	2026-04-09	59	18.6%	0.37	0.60
FTC	cluster_1	U.S. Mid Cap Equity	2026-04-10	58	13.5%	0.40	0.57
IYC	cluster_1	U.S. Mid Cap Equity	2026-07-06	0	0.0%	0.04	0.53
MTUM	cluster_1	U.S. Mid Cap Equity	2026-04-08	60	23.9%	0.50	0.48
XLY	cluster_1	U.S. Mid Cap Equity	2026-07-06	0	0.0%	0.04	0.33
PDP	cluster_1	U.S. Mid Cap Equity	2026-04-08	60	11.1%	0.62	0.28
QMOM	cluster_1	U.S. Mid Cap Equity	2026-06-30	3	-2.8%	0.87	-0.83
HYMB	cluster_16	Municipal Bond	2026-05-26	27	1.1%	0.00	1.00
CMF	cluster_16	Municipal Bond	2026-05-28	25	0.6%	0.00	1.00
NYF	cluster_16	Municipal Bond	2026-05-28	25	0.6%	0.00	1.00
MUB	cluster_16	Municipal Bond	2026-05-28	25	0.5%	0.00	1.00
VTEB	cluster_16	Municipal Bond	2026-05-28	25	0.5%	0.00	1.00
PZA	cluster_16	Municipal Bond	2026-05-29	24	1.2%	0.01	0.99
TFI	cluster_16	Municipal Bond	2026-05-28	25	0.4%	0.01	0.99
HYD	cluster_16	Municipal Bond	2026-05-22	28	1.6%	0.03	0.97
MUNI	cluster_16	Municipal Bond	2026-05-28	25	0.2%	0.03	0.97
FMB	cluster_16	Municipal Bond	2026-05-29	24	0.6%	0.15	0.85
SHM	cluster_16	Municipal Bond	2026-05-28	25	0.1%	0.52	0.48
ITM	cluster_16	Municipal Bond	2026-05-28	25	0.1%	0.68	0.30
CGW	cluster_2	U.S. Large Cap Value	2026-06-25	6	1.0%	0.00	1.00
SMMV	cluster_2	U.S. Large Cap Value	2026-06-25	6	3.1%	0.00	1.00
PHO	cluster_2	U.S. Large Cap Value	2026-06-26	5	0.8%	0.00	1.00
PKW	cluster_2	U.S. Large Cap Value	2026-05-20	30	5.7%	0.00	1.00
SPLV	cluster_2	U.S. Large Cap Value	2026-06-11	15	3.2%	0.00	1.00
REGL	cluster_2	U.S. Large Cap Value	2026-06-12	14	1.8%	0.00	1.00
LGLV	cluster_2	U.S. Large Cap Value	2026-06-26	5	1.3%	0.00	1.00
FIW	cluster_2	U.S. Large Cap Value	2026-06-16	12	2.4%	0.00	1.00
XMLV	cluster_2	U.S. Large Cap Value	2026-06-25	6	1.3%	0.00	1.00
FVD	cluster_2	U.S. Large Cap Value	2026-06-10	16	2.7%	0.00	1.00
FTCS	cluster_2	U.S. Large Cap Value	2026-06-26	5	3.1%	0.00	1.00

LVHD	cluster_2	U.S. Large Cap Value	2026-06-10	16	2.5%	0.00	1.00
AIVL	cluster_2	U.S. Large Cap Value	2026-04-10	58	9.2%	0.00	1.00
SDY	cluster_2	U.S. Large Cap Value	2026-05-20	30	4.6%	0.00	1.00
DIA	cluster_2	U.S. Large Cap Value	2026-04-14	56	9.2%	0.00	1.00
DGRO	cluster_2	U.S. Large Cap Value	2026-04-13	57	6.9%	0.00	1.00
IWD	cluster_2	U.S. Large Cap Value	2026-04-10	58	11.4%	0.01	0.99
RSP	cluster_2	U.S. Large Cap Value	2026-04-14	56	7.7%	0.01	0.99
SPHD	cluster_2	U.S. Large Cap Value	2026-06-08	18	3.4%	0.02	0.98
VONV	cluster_2	U.S. Large Cap Value	2026-04-10	58	11.4%	0.02	0.98
IWX	cluster_2	U.S. Large Cap Value	2026-04-10	58	12.1%	0.02	0.98
MGV	cluster_2	U.S. Large Cap Value	2026-04-13	57	9.7%	0.03	0.97
USMV	cluster_2	U.S. Large Cap Value	2026-06-29	4	1.8%	0.02	0.97
PWV	cluster_2	U.S. Large Cap Value	2026-04-02	63	11.1%	0.03	0.97
FTA	cluster_2	U.S. Large Cap Value	2026-05-21	29	4.2%	0.04	0.96
VTV	cluster_2	U.S. Large Cap Value	2026-04-13	57	8.3%	0.05	0.95
OUSA	cluster_2	U.S. Large Cap Value	2026-07-01	2	1.7%	0.02	0.95
VIG	cluster_2	U.S. Large Cap Value	2026-04-14	56	6.8%	0.07	0.93
HDV	cluster_2	U.S. Large Cap Value	2026-06-25	6	0.6%	0.07	0.90
PRF	cluster_2	U.S. Large Cap Value	2026-04-09	59	10.1%	0.10	0.90
DTD	cluster_2	U.S. Large Cap Value	2026-04-13	57	5.9%	0.11	0.89
FEX	cluster_2	U.S. Large Cap Value	2026-04-09	59	8.7%	0.18	0.82
RWL	cluster_2	U.S. Large Cap Value	2026-04-10	58	9.6%	0.18	0.82
DLN	cluster_2	U.S. Large Cap Value	2026-04-13	57	6.0%	0.21	0.77
SCHV	cluster_2	U.S. Large Cap Value	2026-04-10	58	9.6%	0.27	0.73
FNDX	cluster_2	U.S. Large Cap Value	2026-04-09	59	8.9%	0.26	0.72
DIV	cluster_2	U.S. Large Cap Value	2026-06-29	4	-0.5%	0.03	0.69
JVAL	cluster_2	U.S. Large Cap Value	2026-04-10	58	14.4%	0.49	0.50
ACWV	cluster_2	U.S. Large Cap Value	2026-07-06	0	0.0%	0.17	0.28
VLUE	cluster_2	U.S. Large Cap Value	2026-04-09	59	26.9%	0.96	-0.11
SCHD	cluster_2	U.S. Large Cap Value	2026-06-26	5	0.5%	0.94	-0.86
PSCC	cluster_18	Consumer Staples	2026-06-10	16	4.6%	0.00	1.00
IDU	cluster_18	Consumer Staples	2026-06-25	6	-1.0%	0.00	1.00
XLU	cluster_18	Consumer Staples	2026-06-25	6	-1.2%	0.00	1.00
FUTY	cluster_18	Consumer Staples	2026-06-25	6	-1.3%	0.00	0.99
VPU	cluster_18	Consumer Staples	2026-06-25	6	-1.3%	0.01	0.99
IYK	cluster_18	Consumer Staples	2026-06-09	17	2.3%	0.03	0.97
KXI	cluster_18	Consumer Staples	2026-07-02	1	-0.9%	0.16	0.52
FSTA	cluster_18	Consumer Staples	2026-06-29	4	-0.2%	0.32	0.24
VDC	cluster_18	Consumer Staples	2026-06-29	4	-0.3%	0.34	0.20
XLP	cluster_18	Consumer Staples	2026-07-02	1	-1.0%	0.42	0.17
RWO	cluster_15	Real Estate	2026-06-25	6	1.2%	0.02	0.98
NURE	cluster_15	Real Estate	2026-04-13	57	13.8%	0.05	0.95
INDS	cluster_15	Real Estate	2026-06-08	18	3.6%	0.09	0.90
IYR	cluster_15	Real Estate	2026-06-25	6	0.1%	0.19	0.72
FREL	cluster_15	Real Estate	2026-06-25	6	0.2%	0.25	0.67

XLRE	cluster_15	Real Estate	2026-06-25	6	-0.7%	0.63	-0.07
HEWJ	cluster_22	Japan Equity	2026-04-08	60	9.5%	0.25	0.74
DBJP	cluster_22	Japan Equity	2026-04-08	60	8.9%	0.36	0.61
IPAC	cluster_22	Japan Equity	2026-06-30	3	2.1%	0.47	-0.01
EWJV	cluster_22	Japan Equity	2026-07-06	0	0.0%	0.65	-0.30
BNDX	cluster_6	Aggregate Fixed Income	2026-05-26	27	-0.0%	0.28	0.71
BOND	cluster_6	Aggregate Fixed Income	2026-06-17	11	0.1%	0.28	0.55
IEF	cluster_6	Aggregate Fixed Income	2026-06-26	5	-0.9%	0.25	0.50
BIV	cluster_6	Aggregate Fixed Income	2026-06-26	5	-0.7%	0.25	0.41
TOTL	cluster_6	Aggregate Fixed Income	2026-06-29	4	-0.6%	0.17	0.41
GOVT	cluster_6	Aggregate Fixed Income	2026-06-25	6	-0.6%	0.35	0.41
IGIB	cluster_6	Aggregate Fixed Income	2026-06-25	6	-0.5%	0.28	0.37
LDUR	cluster_6	Aggregate Fixed Income	2026-06-25	6	-0.2%	0.24	0.37
VGIT	cluster_6	Aggregate Fixed Income	2026-06-29	4	-0.6%	0.21	0.37
VCIT	cluster_6	Aggregate Fixed Income	2026-06-25	6	-0.5%	0.29	0.35
SPTI	cluster_6	Aggregate Fixed Income	2026-06-29	4	-0.6%	0.22	0.35
CMBS	cluster_6	Aggregate Fixed Income	2026-06-29	4	-0.3%	0.18	0.34
AGG	cluster_6	Aggregate Fixed Income	2026-06-24	7	-0.5%	0.41	0.32
SPAB	cluster_6	Aggregate Fixed Income	2026-06-24	7	-0.5%	0.40	0.32
BND	cluster_6	Aggregate Fixed Income	2026-06-24	7	-0.6%	0.42	0.30
AGZ	cluster_6	Aggregate Fixed Income	2026-06-26	5	-0.4%	0.34	0.29
GNMA	cluster_6	Aggregate Fixed Income	2026-06-25	6	-0.7%	0.36	0.27
GVI	cluster_6	Aggregate Fixed Income	2026-06-29	4	-0.4%	0.24	0.27
IUSB	cluster_6	Aggregate Fixed Income	2026-06-17	11	0.0%	0.45	0.26
SPIB	cluster_6	Aggregate Fixed Income	2026-06-26	5	-0.4%	0.31	0.23
SCHR	cluster_6	Aggregate Fixed Income	2026-06-29	4	-0.6%	0.27	0.22
SPMB	cluster_6	Aggregate Fixed Income	2026-06-16	12	-0.4%	0.50	0.21
SCHZ	cluster_6	Aggregate Fixed Income	2026-06-24	7	-0.6%	0.46	0.21
VMBS	cluster_6	Aggregate Fixed Income	2026-06-24	7	-0.6%	0.46	0.19
IEI	cluster_6	Aggregate Fixed Income	2026-06-29	4	-0.5%	0.24	0.19
MBB	cluster_6	Aggregate Fixed Income	2026-06-24	7	-0.5%	0.51	0.11
VCSH	cluster_6	Aggregate Fixed Income	2026-06-26	5	-0.3%	0.30	0.10
IAGG	cluster_6	Aggregate Fixed Income	2026-05-22	28	0.1%	0.86	0.06
SLQD	cluster_6	Aggregate Fixed Income	2026-06-26	5	-0.3%	0.31	0.05
AGGY	cluster_6	Aggregate Fixed Income	2026-06-16	12	-0.4%	0.64	0.04
IGSB	cluster_6	Aggregate Fixed Income	2026-06-29	4	-0.4%	0.34	0.01
SPSB	cluster_6	Aggregate Fixed Income	2026-06-26	5	-0.2%	0.36	-0.06
BSV	cluster_6	Aggregate Fixed Income	2026-06-30	3	-0.2%	0.30	-0.07
GTO	cluster_6	Aggregate Fixed Income	2026-06-29	4	-0.3%	0.61	-0.08
ISTB	cluster_6	Aggregate Fixed Income	2026-06-29	4	-0.3%	0.39	-0.15
FBND	cluster_6	Aggregate Fixed Income	2026-06-17	11	-0.0%	0.71	-0.27
SPHQ	cluster_5	U.S. Large Cap Equity	2026-04-10	58	12.4%	0.08	0.92
SPYX	cluster_5	U.S. Large Cap Equity	2026-06-30	3	0.9%	0.27	0.21
URTH	cluster_5	U.S. Large Cap Equity	2026-07-01	2	1.2%	0.27	0.18
GSLC	cluster_5	U.S. Large Cap Equity	2026-06-30	3	0.6%	0.30	0.16

IVV	cluster_5	U.S. Large Cap Equity	2026-06-30	3	0.8%	0.29	0.12
SPY	cluster_5	U.S. Large Cap Equity	2026-06-30	3	0.6%	0.30	0.11
VOO	cluster_5	U.S. Large Cap Equity	2026-06-30	3	0.6%	0.33	0.07
IWL	cluster_5	U.S. Large Cap Equity	2026-07-01	2	0.8%	0.29	0.02
CRBN	cluster_5	U.S. Large Cap Equity	2026-06-30	3	0.6%	0.42	-0.06
MGC	cluster_5	U.S. Large Cap Equity	2026-07-01	2	0.8%	0.35	-0.06
OEF	cluster_5	U.S. Large Cap Equity	2026-07-06	0	0.0%	0.40	-0.30
CATH	cluster_5	U.S. Large Cap Equity	2026-07-06	0	0.0%	0.48	-0.33
DGRW	cluster_5	U.S. Large Cap Equity	2026-07-06	0	0.0%	0.62	-0.50
BAB	cluster_14	Fixed Income	2026-06-16	12	-0.2%	0.36	0.63
TLH	cluster_14	Fixed Income	2026-06-16	12	-0.6%	0.46	0.46
SPTL	cluster_14	Fixed Income	2026-06-16	12	-0.8%	0.55	0.40
VGLT	cluster_14	Fixed Income	2026-06-16	12	-0.8%	0.57	0.36
TLT	cluster_14	Fixed Income	2026-06-16	12	-0.9%	0.64	0.29
USIG	cluster_14	Fixed Income	2026-06-16	12	-0.4%	0.45	0.25
BLV	cluster_14	Fixed Income	2026-06-15	13	-0.4%	0.66	0.23
CORP	cluster_14	Fixed Income	2026-06-16	12	-0.4%	0.44	0.21
FLCO	cluster_14	Fixed Income	2026-06-16	12	-0.4%	0.55	0.14
LQD	cluster_14	Fixed Income	2026-06-16	12	-0.4%	0.58	0.12
IGLB	cluster_14	Fixed Income	2026-06-12	14	-0.5%	0.73	0.08
QLTA	cluster_14	Fixed Income	2026-06-17	11	-0.1%	0.56	0.04
VCLT	cluster_14	Fixed Income	2026-06-12	14	-0.6%	0.77	-0.03
EDV	cluster_14	Fixed Income	2026-06-12	14	-1.5%	0.89	-0.03
SPLB	cluster_14	Fixed Income	2026-06-12	14	-0.6%	0.80	-0.05
IBUY	cluster_3	U.S. Large Cap Growth	2026-06-15	13	3.2%	0.00	1.00
ARKK	cluster_3	U.S. Large Cap Growth	2026-06-29	4	3.7%	0.01	0.95
PSCT	cluster_3	U.S. Large Cap Growth	2026-04-07	61	36.4%	0.14	0.85
QQEW	cluster_3	U.S. Large Cap Growth	2026-04-16	54	18.2%	0.19	0.81
IPO	cluster_3	U.S. Large Cap Growth	2026-04-13	57	26.9%	0.20	0.80
PSI	cluster_3	U.S. Large Cap Growth	2026-04-07	61	60.3%	0.21	0.79
SPHB	cluster_3	U.S. Large Cap Growth	2026-04-10	58	22.9%	0.49	0.46
XNTK	cluster_3	U.S. Large Cap Growth	2026-04-10	58	34.1%	0.55	0.40
QTEC	cluster_3	U.S. Large Cap Growth	2026-04-10	58	41.7%	0.55	0.37
ARKW	cluster_3	U.S. Large Cap Growth	2026-07-02	1	3.3%	0.13	0.37
SOXX	cluster_3	U.S. Large Cap Growth	2026-04-08	60	57.0%	0.68	0.27
SMH	cluster_3	U.S. Large Cap Growth	2026-04-08	60	42.9%	0.67	0.21
SCHG	cluster_3	U.S. Large Cap Growth	2026-07-06	0	0.0%	0.18	0.06
IUSG	cluster_3	U.S. Large Cap Growth	2026-07-01	2	0.1%	0.36	-0.09
FTXL	cluster_3	U.S. Large Cap Growth	2026-04-07	61	63.3%	0.84	-0.10
IVW	cluster_3	U.S. Large Cap Growth	2026-07-01	2	0.2%	0.37	-0.12
SPYG	cluster_3	U.S. Large Cap Growth	2026-07-01	2	0.2%	0.38	-0.14
VOOG	cluster_3	U.S. Large Cap Growth	2026-07-01	2	0.2%	0.38	-0.14
QTUM	cluster_3	U.S. Large Cap Growth	2026-04-10	58	34.5%	0.88	-0.16
ROBO	cluster_3	U.S. Large Cap Growth	2026-07-06	0	0.0%	0.35	-0.19
ILCG	cluster_3	U.S. Large Cap Growth	2026-06-30	3	-1.2%	0.45	-0.24

MGK	cluster_3	U.S. Large Cap Growth	2026-07-06	0	0.0%	0.43	-0.31
ONEQ	cluster_3	U.S. Large Cap Growth	2026-07-06	0	0.0%	0.48	-0.36
SKYY	cluster_3	U.S. Large Cap Growth	2026-07-01	2	0.9%	0.67	-0.41
ARKQ	cluster_3	U.S. Large Cap Growth	2026-06-17	11	-1.0%	0.52	-0.47
XSD	cluster_3	U.S. Large Cap Growth	2026-04-09	59	56.8%	0.90	-0.70
PRNT	cluster_3	U.S. Large Cap Growth	2026-06-30	3	0.3%	0.81	-0.72
PTF	cluster_3	U.S. Large Cap Growth	2026-04-07	61	24.3%	0.96	-0.79
EUFN	cluster_4	Developed ex-US Equity	2026-06-12	14	4.2%	0.00	1.00
IHDG	cluster_4	Developed ex-US Equity	2026-05-06	40	5.0%	0.12	0.88
VIGI	cluster_4	Developed ex-US Equity	2026-06-30	3	2.3%	0.07	0.86
IQLT	cluster_4	Developed ex-US Equity	2026-07-01	2	2.0%	0.07	0.75
IEV	cluster_4	Developed ex-US Equity	2026-07-01	2	2.4%	0.12	0.73
EWO	cluster_4	Developed ex-US Equity	2026-04-10	58	11.9%	0.29	0.70
QEFA	cluster_4	Developed ex-US Equity	2026-07-06	0	0.0%	0.03	0.57
ESGD	cluster_4	Developed ex-US Equity	2026-06-30	3	1.6%	0.22	0.49
IEUR	cluster_4	Developed ex-US Equity	2026-07-02	1	0.7%	0.19	0.46
VGK	cluster_4	Developed ex-US Equity	2026-07-01	2	2.5%	0.30	0.45
EWN	cluster_4	Developed ex-US Equity	2026-04-10	58	10.2%	0.55	0.44
EFA	cluster_4	Developed ex-US Equity	2026-07-01	2	2.4%	0.28	0.32
GSIE	cluster_4	Developed ex-US Equity	2026-07-02	1	1.0%	0.40	0.24
IEFA	cluster_4	Developed ex-US Equity	2026-07-06	0	0.0%	0.34	0.10
EWQ	cluster_4	Developed ex-US Equity	2026-07-06	0	0.0%	0.68	-0.32
IXUS	cluster_4	Developed ex-US Equity	2026-07-06	0	0.0%	0.67	-0.46
SPDW	cluster_4	Developed ex-US Equity	2026-06-30	3	0.9%	0.77	-0.50
IQDF	cluster_4	Developed ex-US Equity	2026-07-06	0	0.0%	0.77	-0.55
UUP	cluster_26	Leveraged / Inverse	2026-05-18	32	2.2%	0.03	0.97
NEAR	cluster_24	Cash / Ultra-Short Fixe...	2026-06-30	3	-0.2%	0.43	-0.33
EWL	cluster_8	Asia-Pacific Equity	2026-06-25	6	1.4%	0.00	1.00
EWS	cluster_8	Asia-Pacific Equity	2026-06-17	11	2.3%	0.03	0.95
IGF	cluster_8	Asia-Pacific Equity	2026-06-29	4	-0.5%	0.11	0.35
EMLC	cluster_8	Asia-Pacific Equity	2026-06-30	3	0.1%	0.34	0.29
EBND	cluster_8	Asia-Pacific Equity	2026-06-17	11	-0.4%	0.42	0.01
KURE	cluster_13	China Equity	2026-07-06	0	0.0%	0.00	0.96
ASHR	cluster_13	China Equity	2026-06-15	13	-1.5%	0.26	0.32
CNYA	cluster_13	China Equity	2026-06-15	13	-2.2%	0.25	0.26
EWT	cluster_9	Emerging Markets Equity	2026-04-08	60	40.5%	0.39	0.59
SCHE	cluster_9	Emerging Markets Equity	2026-07-06	0	0.0%	0.53	-0.27
VWO	cluster_9	Emerging Markets Equity	2026-07-06	0	0.0%	0.54	-0.30
EWX	cluster_9	Emerging Markets Equity	2026-06-15	13	-1.2%	0.59	-0.40
IEMG	cluster_9	Emerging Markets Equity	2026-07-06	0	0.0%	0.80	-0.65
FALN	cluster_11	High Yield Bond	2026-06-12	14	0.4%	0.03	0.96
ANGL	cluster_11	High Yield Bond	2026-06-15	13	0.0%	0.10	0.83
VWOB	cluster_11	High Yield Bond	2026-05-26	27	0.6%	0.53	0.34
VRP	cluster_11	High Yield Bond	2026-07-02	1	0.0%	0.30	0.29
PCY	cluster_11	High Yield Bond	2026-05-28	25	0.7%	0.71	0.18

BBHY	cluster_11	High Yield Bond	2026-06-25	6	-0.1%	0.37	-0.05
HYG	cluster_11	High Yield Bond	2026-06-30	3	-0.1%	0.45	-0.22
HYLB	cluster_11	High Yield Bond	2026-06-30	3	-0.2%	0.44	-0.25
JNK	cluster_11	High Yield Bond	2026-06-16	12	-0.3%	0.53	-0.31
SHYG	cluster_11	High Yield Bond	2026-06-30	3	-0.3%	0.54	-0.45
SJNK	cluster_11	High Yield Bond	2026-06-17	11	-0.0%	0.56	-0.47
HYS	cluster_11	High Yield Bond	2026-06-16	12	-0.3%	0.73	-0.63
SIVR	cluster_17	Gold	2026-05-27	26	-16.8%	0.97	-0.97
SLV	cluster_17	Gold	2026-05-27	26	-16.9%	0.97	-0.97
EMLP	cluster_10	Materials	2026-06-26	5	-1.5%	0.07	0.50
XLB	cluster_10	Materials	2026-07-06	0	0.0%	0.62	-0.14
MLPA	cluster_10	Materials	2026-06-08	18	-1.6%	0.24	-0.23
MLPX	cluster_10	Materials	2026-06-26	5	-3.0%	0.41	-0.28
RSPM	cluster_10	Materials	2026-06-16	12	-1.9%	0.73	-0.45
VAW	cluster_10	Materials	2026-06-16	12	-2.5%	0.82	-0.63
DJP	cluster_19	Broad Commodities	2026-06-03	21	-9.0%	1.00	-1.00
DBE	cluster_19	Broad Commodities	2026-06-03	21	-18.6%	1.00	-1.00
DBC	cluster_19	Broad Commodities	2026-06-03	21	-10.9%	1.00	-1.00
PDBC	cluster_19	Broad Commodities	2026-06-03	21	-10.8%	1.00	-1.00
USO	cluster_19	Broad Commodities	2026-06-03	21	-25.9%	1.00	-1.00

ETF Net Signal Strength

NSS = P(buy within h=5 days) - P(sell within h=5 days), range [-1, 1]. ETF rows only, cluster-deduped (top 1 ETF per cluster) so each row is a distinct theme. Bullish side filters to ETFs whose cluster has positive trailing return – i.e. would not be blocked by the entry filter.

Top 20 most bullish (cluster-deduped, abs-mom approved)

ticker	cluster	theme	p_buy	nss
IBB	cluster_12	Healthcare	1.00	1.00
KIE	cluster_7	U.S. Small Cap Equity	1.00	1.00
CGW	cluster_2	U.S. Large Cap Value	1.00	1.00
IJT	cluster_1	U.S. Mid Cap Equity	1.00	1.00
REZ	cluster_15	Real Estate	1.00	1.00
HYMB	cluster_16	Municipal Bond	1.00	1.00
IBUY	cluster_3	U.S. Large Cap Growth	1.00	1.00
PSCC	cluster_18	Consumer Staples	1.00	1.00
EWP	cluster_4	Developed ex-US Equity	1.00	1.00
CQQQ	cluster_13	China Equity	1.00	1.00
EWL	cluster_8	Asia-Pacific Equity	1.00	1.00
QUS	cluster_5	U.S. Large Cap Equity	0.99	0.99
FALN	cluster_11	High Yield Bond	0.99	0.96
DXJ	cluster_22	Japan Equity	1.00	0.87
FLTR	cluster_23	Bank Loan / Floating Rate	0.98	0.83
BNDX	cluster_6	Aggregate Fixed Income	0.99	0.71
BAB	cluster_14	Fixed Income	0.99	0.63
MOO	cluster_10	Materials	0.62	0.62
EWI	cluster_9	Emerging Markets Equity	0.98	0.59
MINT	cluster_24	Cash / Ultra-Short Fixed ...	0.81	0.39

Top 20 most bearish (cluster-deduped)

ticker	cluster	theme	p_sell	nss
NORW	cluster_10	Materials	1.00	-1.00
USO	cluster_19	Broad Commodities	1.00	-1.00
VTIP	cluster_21	Inflation-Protected Fixed...	1.00	-1.00
FXI	cluster_13	China Equity	1.00	-1.00
FXI	cluster_25	International Fixed Income	1.00	-1.00
PFXF	cluster_11	High Yield Bond	1.00	-1.00

DVYE	cluster_9	Emerging Markets Equity	1.00	-1.00
FXC	cluster_8	Asia-Pacific Equity	1.00	-1.00
IDV	cluster_4	Developed ex-US Equity	1.00	-1.00
BLCN	cluster_3	U.S. Large Cap Growth	1.00	-1.00
IGBH	cluster_23	Bank Loan / Floating Rate	1.00	-0.99
FDL	cluster_2	U.S. Large Cap Value	0.99	-0.97
SLV	cluster_17	Gold	0.97	-0.97
EWM	cluster_20	India Equity	0.92	-0.92
FTSM	cluster_24	Cash / Ultra-Short Fixed ...	0.87	-0.85
QMOM	cluster_1	U.S. Mid Cap Equity	0.87	-0.83
LMBS	cluster_6	Aggregate Fixed Income	0.88	-0.82
PSP	cluster_7	U.S. Small Cap Equity	0.79	-0.79
SUB	cluster_16	Municipal Bond	1.00	-0.77
I00	cluster_5	U.S. Large Cap Equity	0.69	-0.69

Diversifier Candidates

Top ETFs ranked by lowest composite of (max correlation against any peer, |PC1 loading|). Both small means the ETF moves uniquely – candidates for portfolio diversification regardless of which cluster they sit in.

ticker	name	category	cluster_id	cluster_size	etf_signal	max_corr_to_others	median_corr_to_others	abs_pc1_loading	compo
PPLT	abrdn Physical Platinum Shares ...	Platinum	cluster_20	10	FLAT	0.306	0.095	0.00021	
UNG	United States Natural Gas Fund ...	Natural Gas	cluster_19	11	FLAT	0.485	0.076	0.00017	
GBIL	Goldman Sachs Access Treasury 0...	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.741	0.037	0.00008	
FXV	Invesco CurrencyShares Japanese...	Currency	cluster_25	6	FLAT	0.730	0.094	0.00022	
VNLA	Janus Henderson Short Duration ...	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.634	0.214	0.00035	
DBA	Invesco DB Agriculture Fund	Agricultural Commodit...	cluster_19	11	FLAT	0.596	0.170	0.00040	
FTSM	First Trust Enhanced Short Matu...	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.742	0.125	0.00030	
ICSH	iShares Ultra Short Duration Bo...	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.742	0.155	0.00039	
MINT	PIMCO Enhanced Short Maturity A...	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.679	0.179	0.00046	
NEAR	iShares Short Duration Bond Act...	Cash / Ultra-Short Fi...	cluster_24	9	LONG	0.697	0.191	0.00045	
GSY	Invesco Ultra Short Duration ETF	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.731	0.180	0.00044	
TUR	iShares MSCI Turkey ETF	Turkey Equity	cluster_20	10	FLAT	0.501	0.219	0.00056	
LDUR	PIMCO Enhanced Low Duration Act...	Fixed Income	cluster_6	43	LONG	0.847	0.127	0.00036	
BIL	State Street SPDR Bloomberg 1-3...	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.870	0.020	0.00001	
SHV	iShares 0-1 Year Treasury Bond ...	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.870	0.075	0.00021	
CMBS	iShares CMBS Bond ETF	Mortgage-Backed Fixed...	cluster_6	43	LONG	0.876	0.160	0.00031	
FXF	Invesco CurrencyShares Swiss Fr...	Currency	cluster_25	6	FLAT	0.820	0.204	0.00049	
LMBS	First Trust Low Duration Opport...	Mortgage-Backed Fixed...	cluster_6	43	FLAT	0.835	0.237	0.00047	
NOBL	ProShares S&P 500 Dividend Aris...	U.S. Dividend Equity	cluster_20	10	FLAT	0.446	0.274	0.00075	
SHM	State Street SPDR Nuveen ICE Sh...	Municipal Bond	cluster_16	13	LONG	0.877	0.248	0.00046	
VNM	VanEck Vietnam ETF	Vietnam Equity	cluster_20	10	FLAT	0.444	0.328	0.00083	
SUB	iShares Short-Term National Mun...	Municipal Bond	cluster_16	13	FLAT	0.877	0.263	0.00054	
KURE	KraneShares MSCI All China Heal...	China Equity	cluster_13	15	LONG	0.794	0.270	0.00071	
DBB	Invesco DB Base Metals Fund	Industrial Metals	cluster_10	21	FLAT	0.749	0.360	0.00085	
EIDO	iShares MSCI Indonesia ETF	Indonesia Equity	cluster_20	10	FLAT	0.505	0.355	0.00102	
FLTR	VanEck IG Floating Rate ETF	Floating Rate Fixed I...	cluster_23	9	FLAT	0.866	0.325	0.00071	
EPHE	iShares MSCI Philippines ETF	Philippines Equity	cluster_20	10	FLAT	0.562	0.408	0.00107	
KBA	KraneShares Bosera MSCI China A...	China Equity	cluster_13	15	FLAT	0.857	0.257	0.00073	
CNXT	VanEck ChiNext Innovators ETF	China Equity	cluster_13	15	FLAT	0.903	0.240	0.00064	
LTPZ	Pimco 15 Year U.S. TIPS Index E...	Inflation-Protected F...	cluster_21	10	FLAT	0.918	0.226	0.00057	

Run Parameters

Screen params (signal construction) and backtest params (sizing + strategy filters) shown together. The backtest block is populated by `scripts/04_run_backtest.R` and may be absent on screen-only runs.

	scope	parameter	value
short_ma	screen	short_ma	5
buy_ma	screen	buy_ma	50
sell_ma	screen	sell_ma	20
vol_window	screen	vol_window	20
horizon	screen	horizon	5
imminent_horizon	screen	imminent_horizon	1
cluster_smoothing_window	screen	cluster_smoothing_window	3
sell_cushion_k	screen	sell_cushion_k	0.5
sell_cushion_sd_window	screen	sell_cushion_sd_window	20